



PRINCE SATTAM BIN ABDULAZIZ UNIVERSITY
Graduate Studies
Digital Transformation and Emerging Technologies Program

**An Integrated Readiness Framework for Assessing AI-Enabled Digital
Transformation and Emerging Technologies
in Saudi Smart-City Megaprojects: A NEOM Case Study**

إطار جاهزية متكامل لتقييم التحول الرقمي المدعوم بالذكاء الاصطناعي والتقنيات الناشئة
في مشاريع المدن الذكية الكبرى في المملكة العربية السعودية: دراسة حالة نيوم

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ABSTRACT

Saudi Arabia's shift toward data-driven public services and smart-city development has made AI governance a managerial problem, not merely a technical one. Managers now need to decide whether AI-enabled transformation initiatives are strategically aligned, financially disciplined, operationally realistic, governable, trusted, and ethically defensible. Yet the existing literature too often treats these dimensions as separate conversations.

This proposal develops and evaluates an integrated digital-transformation readiness framework for AI-enabled digital transformation in Saudi smart-city megaprojects, using NEOM as a bounded conceptual case setting. It compares a generic digital-transformation checklist with a framework that combines strategic alignment, ROI discipline, operational feasibility, data governance, stakeholder trust, and Islamic business-ethics criteria. The methodology combines thematic literature synthesis, directed document analysis, codebook construction, pilot-tested scoring, synthetic NEOM-based scenarios, paired comparison, coverage analysis, and sensitivity testing. The expected contribution is a reproducible managerial assessment tool, not a technical AI system or an operational audit of NEOM. The project deliberately avoids confidential NEOM data and real implementation claims.

Word count: approximately 169 words.

KEYWORDS

Digital Transformation and Emerging Technologies, digital transformation, AI governance, strategic performance readiness, NEOM, Saudi Arabia, Vision 2030, ROI, stakeholder trust, data governance, Islamic business ethics, smart cities

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LIST OF ACRONYMS AND ABBREVIATIONS

Acronym	Meaning
AI	Artificial Intelligence
DTET	Master in Digital Transformation and Emerging Technologies
MCDA	Multi-Criteria Decision Analysis
PDPL	Personal Data Protection Law
ROI	Return on Investment
CAPEX	Capital Expenditure
OPEX	Operating Expenditure
SDAIA	Saudi Data and Artificial Intelligence Authority
KPI	Key Performance Indicator
WP	Work Package
PSAU	Prince Sattam bin Abdulaziz University

DEFINITION OF KEY CONSTRUCTS

The following operational definitions set the boundaries of the proposal before the research problem, research questions, and methodology appear. They clarify how the project classifies evidence, builds the evaluation rubric, and interprets synthetic business decision scenarios.

Table K1 anchors the study's key constructs before the research problem and methodology are introduced.

Table K1. Operational Definitions of Key Constructs

Construct / term	Operational definition used in this study	Source / authority
AI-enabled digital transformation governance	A managerial control system consisting of strategy alignment, data governance, investment discipline, risk controls, accountability, documentation, and monitoring procedures used to guide AI-enabled transformation initiatives.	Synthesized from digital transformation, AI governance, and strategic management literature.
Strategic performance readiness	The dependent response variable: the degree to which a proposed AI-enabled initiative is prepared to generate measurable strategic value while controlling operational, financial, regulatory, and stakeholder risks.	Novel operational construct proposed in this study.
Business value realization	The conversion of digital initiatives into measurable organizational outcomes such as efficiency, cost control, revenue potential, service quality, or decision speed.	Synthesized from digital business strategy and performance-management literature.
Governance risk	The probability that unclear accountability, weak data controls, poor documentation, or misaligned incentives reduce the strategic value of a digital initiative.	Synthesized from AI-risk and corporate-governance literature.
Stakeholder trust	The perceived legitimacy of AI-enabled business decisions by users, regulators, managers, partners, and the wider public.	Synthesized from stakeholder and responsible-AI literature.
Al-Amanah	Trustworthy stewardship of data, resources, financial assumptions, and managerial accountability.	Operationalized from Islamic business-ethics and AI-ethics literature.
Ihsan	Continuous improvement, disciplined monitoring, service excellence, and correction of avoidable errors.	Operationalized from Islamic business-ethics and quality-management logic.
Maslaha	Public interest, harm reduction, service accessibility, fairness, and social benefit in business decisions.	Operationalized from Islamic ethics and stakeholder-value logic.

Note. AI = artificial intelligence; DTET = Digital Transformation and Emerging Technologies. The definitions are operational definitions used only for this proposal and are not intended as universal definitions.

These definitions reduce conceptual drift. Broader issues such as national competitiveness, total real-estate performance, technical model engineering, and confidential operational performance inside NEOM remain outside this digital-transformation project.

1. INTRODUCTION

AI-enabled digital transformation has moved beyond the IT department. It is now a strategic management problem. Large organizations no longer adopt artificial intelligence only to automate isolated tasks; they use it to redesign services, allocate resources, monitor performance, manage risk, and defend investment decisions (Davenport & Ronanki, 2018). In Saudi Arabia, this matters because Vision 2030 links national transformation to diversification, private-sector growth, technology adoption, and stronger institutional performance (Saudi Vision 2030, n.d.). At the same time, the national data and AI agenda, personal-data protection requirements, and large smart-city projects increase the pressure on managers to evaluate digital transformation through strategy, governance, performance, and public value, not through technical feasibility alone (Saudi Data and Artificial Intelligence Authority, n.d.-a; Saudi Data and Artificial Intelligence Authority, n.d.-c).

The literature already shows that technology creates value only when organizations connect it to capabilities, governance structures, business models, and performance measurement (Bharadwaj et al., 2013; Verhoef et al., 2021; Vial, 2019). Strategic management research adds a second point: resources matter only when managers deploy them coherently and turn them into sustainable performance (Barney, 1991; Porter, 1985; Teece et al., 1997). AI governance frameworks add a third layer by stressing accountability, transparency, risk management, human oversight, and responsible deployment (National Institute of Standards and Technology, 2023; Organisation for Economic Co-operation and Development, n.d.). The problem is not that these streams are weak. The problem is that they remain fragmented for DTET-level decision-making. Maturity models often stop at technological adoption. Governance frameworks often remain principle-based. Strategy tools often focus on competitive positioning. What is missing is a measurable framework that helps managers judge whether an AI-enabled initiative is strategically aligned, financially disciplined, operationally feasible, governable, and ethically credible in a Saudi smart-city megaproject context.

That gap matters because smart-city megaprojects force high-stakes managerial decisions. They involve capital-intensive infrastructure, complex stakeholders, data-dependent services, public-value expectations, and reputational risk. NEOM offers a useful bounded case setting because it brings these pressures together in a visible Saudi smart-city megaproject (NEOM, n.d.). This proposal does not evaluate NEOM as a whole, measure its real performance, access confidential institutional data, or claim to assess actual implementation outcomes. It uses NEOM as a conceptual case setting to ask a narrower business-administration question: how can managers evaluate the readiness of AI-enabled transformation initiatives before treating them as strategically sound?

The analytical model is deliberately simple. The independent variable is assessment framework type or integration level: a generic digital-transformation checklist versus the proposed integrated digital-transformation-oriented readiness framework. The dependent variable is strategic performance-readiness assessment quality, meaning the extent to which the assessment captures the strategic, financial, operational, governance, stakeholder-trust, and Islamic-ethical dimensions required for managerial decision-making. The assessment dimensions include strategic alignment, investment discipline, operational feasibility, governance-risk control, stakeholder trust, and ethical legitimacy.

On that basis, this digital-transformation research develops and evaluates an integrated strategic performance-readiness framework for AI-enabled digital transformation initiatives in Saudi smart-city megaprojects. The framework will draw on document analysis and literature synthesis, translate the evidence into measurable variables and scoring criteria, and test the model through synthetic NEOM-based business decision scenarios. Its contribution is not to design an AI system or certify a real megaproject. Its contribution is to offer a bounded, reproducible managerial assessment tool that helps graduate-level decision-makers test whether integrated readiness criteria add analytical coverage beyond a generic digital-transformation checklist.

2. RESEARCH BACKGROUND

This proposal belongs in digital transformation and emerging technologies because it examines how managers govern digital transformation initiatives, allocate resources, define performance criteria, and control strategic risk. AI remains technically relevant, but here it is treated as a managerial capability embedded in organizational strategy. The business question is sharper: how can managers evaluate value creation, investment discipline, operational effectiveness, stakeholder trust, and governance accountability before approving AI-enabled transformation?

Four background trends shape the proposal. First, digital transformation research shows that technology creates value only when it connects to strategy, organizational capability, and business-model redesign. Second, AI governance literature shows how weak documentation, unclear accountability, and low transparency can undermine trust and performance. Third, performance-management research shows why financial metrics must be read alongside operational and stakeholder indicators. Fourth, Saudi policy and smart-city development create a context in which technology adoption must align with regulation, national transformation goals, and culturally grounded ethics.

For that reason, the proposal uses NEOM as a case setting for a strategic-management question: how can graduate-level managers judge whether AI-enabled transformation initiatives are ready to produce risk-adjusted business value? The answer proposed here is a structured readiness framework that brings strategy, governance, finance, operations, risk, and ethics into one auditable decision model.

3. RESEARCH PROBLEM AND RESEARCH QUESTIONS

3.1 Problem Statement

AI-enabled digital transformation in Saudi smart-city megaprojects creates a demanding managerial evaluation problem. Managers must judge whether proposed initiatives are strategically aligned, financially disciplined, operationally feasible, governable, trusted by stakeholders, and ethically defensible. Existing approaches do not fully answer that question. Digital-transformation maturity models often emphasize technological adoption; AI-governance frameworks emphasize risk and accountability; business-case tools emphasize ROI; and ethical discussions often remain normative instead of operational. These dimensions are therefore assessed separately, leaving a clear research gap: there is limited evidence on whether an integrated digital-transformation-oriented readiness framework can assess strategic performance readiness more comprehensively than a generic digital-transformation checklist.

This project does not try to audit NEOM, access confidential data, or judge the success of Saudi megaprojects. It uses NEOM as a bounded conceptual case setting and relies on public documents, academic literature, coded criteria, and synthetic business decision scenarios. The research problem is not how to “fix” digital transformation in smart cities. It is how to determine whether a structured, integrated assessment framework gives managers stronger analytical coverage under controlled digital-transformation research conditions.

3.2 Main Research Question

To what extent can an integrated digital-transformation-oriented readiness framework improve the assessment of strategic performance readiness for AI-enabled digital transformation initiatives in Saudi smart-city megaprojects, compared with a generic digital-transformation checklist?

3.3 Specific Sub-questions

- RQ1. What strategic, financial, operational, governance, stakeholder-trust, and ethical requirements define strategic performance readiness for AI-enabled digital transformation initiatives in Saudi smart-city megaprojects?
- RQ2. How can these requirements be converted into measurable variables and explicit scoring criteria for an digital-transformation-oriented readiness model?
- RQ3. How can the Islamic business-ethics principles of Al-Amanah, Ihsan, and Maslaha be translated into auditable managerial criteria without replacing financial, operational, or governance controls?
- RQ4. To what extent does the integrated framework provide additional analytical coverage compared with a generic digital-transformation checklist when applied to synthetic NEOM-based business decision scenarios?

3.4 Delimitation and Variable Boundaries

The project draws firm boundaries around its variables. The independent variable is assessment framework type or integration level. The dependent variable is strategic performance-readiness assessment quality. The assessment dimensions are strategic alignment, investment discipline, operational feasibility, governance-risk control, stakeholder trust, and Islamic ethical legitimacy. Control conditions include the NEOM conceptual case setting, the public and non-confidential source base, the synthetic scenario design, the scoring scale, and the absence of real implementation data. Other possible variables—macroeconomic impact, citizen adoption, procurement execution, and actual NEOM financial performance—are acknowledged but excluded from this DTET scope.

3.5 Quantified Problem Funnel

Table 1 narrows the research problem from the national transformation context to the specific assessment gap addressed by this study.

Table 1. Problem Funnel and Scope Boundaries for the Research Gap

Funnel level	Empirical marker	Study implication
National transformation	Vision 2030 emphasizes diversification, private-sector development, technology-enabled competitiveness, and improved public services.	Digital transformation should be assessed not only for technical ambition but also for strategic and economic legitimacy.
Digital Transformation and Emerging Technologies program context	Study in Saudi lists Digital Transformation and Emerging Technologies at master's level at Prince Sattam bin Abdulaziz University for international students, with Arabic and English language options.	The proposal must frame the topic as an applied digital-transformation and emerging-technologies research problem, not as a narrow computer-science thesis.
Institutional fit	Prince Sattam bin Abdulaziz University's graduate-program environment includes digital-transformation, emerging-technology, IT, cybersecurity, and quality/development structures.	The framework should focus on managerial decision-making, governance, performance, and value realization.
Digital governance need	AI-enabled transformation initiatives create performance opportunities but also risks in data control, accountability, transparency, and stakeholder trust.	The study needs a readiness framework that integrates finance, operations, governance, and ethics.
Case setting	NEOM provides a visible Saudi smart-city megaproject where digital transformation governance can be studied using public and synthetic evidence.	The case supports a bounded DTET analysis without claiming operational access.
Technical-governance gap	Many digital transformation analyses separate ROI, governance, ethics, and operational readiness.	The study evaluates strategic readiness through an integrated scoring model rather than a generic checklist.

Note. AI = artificial intelligence. The table structures the movement from broad policy context to the specific digital-transformation research gap. It does not measure real NEOM performance.

Existing solutions include digital-transformation maturity models, AI-governance principles, corporate-governance controls, balanced-scorecard logic, and financial business-case methods. The current consensus is that digital transformation can improve organizational performance; the unresolved research issue is how managers

should assess readiness before deployment when public value, data governance, stakeholder trust, and ethical legitimacy matter simultaneously. The unanswered question is whether an integrated DTET framework provides additional analytical coverage beyond a generic digital-transformation checklist under synthetic scenario conditions.

4. RESEARCH AIM AND OBJECTIVES

4.1 Main Aim

The main aim of this digital-transformation research is to develop and evaluate an integrated strategic performance-readiness framework for assessing AI-enabled digital transformation initiatives in Saudi smart-city megaprojects, using NEOM as a bounded case setting.

This aim sets one clear destination. The project will not evaluate NEOM's real performance, access confidential institutional data, or measure national transformation outcomes. It will build and test a managerial assessment framework under controlled research conditions.

4.2 Specific Objectives

The project will pursue four specific, measurable, and sequential objectives:

- To identify and code at least 15-20 official, academic, and digital-transformation and emerging-technologies sources relevant to AI-enabled digital transformation readiness in Saudi smart-city megaprojects by Month 4, using predefined code families covering strategy, financial discipline, operations, governance, stakeholder trust, and Islamic business ethics.
- To operationalize the coded requirements into a measurable digital-transformation-oriented readiness model by Month 5, defining the independent, dependent, mediating, and control variables and translating them into explicit 0-3 scoring criteria.
- To translate and map the Islamic business-ethics principles of Al-Amanah, Ihsan, and Maslaha into auditable managerial criteria by Month 8, then compare their coverage against generic governance indicators to determine whether they add distinct analytical value.
- To evaluate the integrated framework against a generic digital-transformation checklist by Month 10, using at least 30 synthetic NEOM-based business decision scenarios, paired scoring, coverage comparison, and sensitivity analysis.

4.3 Research Questions and SMART Objectives

Table 2 demonstrates that the research design is internally aligned: each research question is matched with one measurable objective, one indicator, and one timeframe.

Table 2. Alignment Between Research Questions, SMART Objectives, Indicators, and Timeframes

Research question	SMART objective	Measurable indicator	Timeframe
RQ1. What strategic, financial, operational, governance, and ethical requirements define strategic performance readiness for AI-enabled digital transformation initiatives in Saudi smart-city megaprojects?	1. To identify and code at least 15-20 official, academic, and digital-transformation and emerging-technologies sources relevant to AI-enabled digital transformation readiness.	A coded source matrix covering at least 15 sources and predefined code families.	Months 1-4
RQ2. How can these requirements be converted into measurable variables and scoring criteria for an digital-transformation-oriented readiness model?	2. To operationalize the coded requirements into independent, dependent, mediating, and control variables with explicit 0-3 scoring criteria.	A variable table, scoring rubric, and operational definitions linked to strategy, ROI, operations, governance, stakeholder trust, and ethics.	Months 4-5

Research question	SMART objective	Measurable indicator	Timeframe
RQ3. How can the Islamic business-ethics principles of Al-Amanah, Ihsan, and Maslaha be translated into auditable managerial criteria without replacing financial, operational, or governance controls?	3. To translate and map Al-Amanah, Ihsan, and Maslaha into measurable audit criteria and compare their coverage against generic governance indicators.	An ethics-by-design checklist with item-level mapping and coverage-gain notes.	Months 6-8
RQ4. To what extent does the integrated framework provide additional analytical coverage compared with a generic digital-transformation checklist when applied to synthetic NEOM-based business decision scenarios?	4. To evaluate the integrated framework against a baseline checklist using at least 30 synthetic NEOM-based business decision scenarios.	A paired score comparison, coverage comparison, and sensitivity-analysis summary showing whether and where the integrated framework adds analytical coverage.	Months 8-10

Note. SMART = specific, measurable, achievable, realistic, and time-constrained. The table is used as an alignment control to prevent unmapped objectives or unsupported research questions.

4.4 Objective-Question-Method-Deliverable Alignment

Table 2A strengthens traceability by linking each objective to the corresponding question, method, data source, and deliverable.

Table 2A. Objective-Question-Method-Deliverable Alignment Matrix

Specific objective	Mirrored research question	Method and data source	Deliverable / evidence of progress
O1. Identify and code Saudi digital-transformation and business-governance requirements by Month 4.	RQ1	Directed content analysis of Vision 2030, Study in Saudi, Prince Sattam bin Abdulaziz University materials, SDAIA/PDPL materials, NEOM public documents, and strategic-management/digital-transformation literature using a predefined codebook.	Coded source matrix, governance requirement map, codebook, and literature-policy synthesis.
O2. Operationalize requirements into an digital-transformation readiness model by Month 5.	RQ2	Conceptual synthesis and variable classification across strategy alignment, ROI discipline, operational feasibility, data governance, accountability, stakeholder trust, and ethics; conversion into 0-3 scoring criteria.	Variable model, operational definitions table, scoring rubric, and conceptual framework figure.
O3. Translate Islamic ethical principles into auditable business criteria by Month 8.	RQ3	Concept-to-control mapping of Al-Amanah, Ihsan, and Maslaha into measurable audit items, followed by coverage comparison against generic governance indicators.	Ethics-by-design checklist, item-level mapping table, and coverage-gain notes.
O4. Evaluate the integrated framework against a baseline checklist by Month 10.	RQ4	Synthetic NEOM-based business decision scenarios; paired scoring; baseline versus integrated-readiness comparison; coverage comparison; sensitivity analysis.	Scenario-analysis chapter, scored comparison table, sensitivity-analysis notes, and reproducibility package.

Note. DTET = Digital Transformation and Emerging Technologies. Each objective is linked to a research question, method, data source, and deliverable to protect methodological traceability.

The alignment table constrains the scope of the proposal: no method is included unless it directly serves one objective, and no objective is retained unless it can be answered with a named analytical procedure, a measurable indicator, and a concrete evidence trail. Final outputs such as the framework,

audit checklist, scenario protocol, and managerial brief are treated as deliverables instead of standalone research objectives.

5. LITERATURE REVIEW

This literature review is organized thematically, not chronologically. It does not summarize authors one by one. It asks a more useful question: how do separate research streams explain, and still fail to explain, the assessment of AI-enabled digital transformation readiness in Saudi smart-city megaprojects? The review brings together six bodies of work: strategic management, digital transformation, AI governance, performance measurement, Saudi institutional context, and Islamic business ethics.

5.1 Strategic Management and the Logic of Readiness

Strategic management provides the first foundation because AI-enabled digital transformation is not merely a technical investment. It is a strategic decision about resources, capabilities, value creation, and organizational positioning. Porter's value-chain logic explains how organizations create advantage through activities, cost discipline, and differentiation (Porter, 1985). Barney's resource-based view shifts the focus from activities to resources by arguing that sustainable advantage depends on resources that are valuable, rare, difficult to imitate, and non-substitutable (Barney, 1991). Teece's dynamic capabilities framework extends the argument by emphasizing how organizations sense opportunities, seize them, and reconfigure resources under changing conditions (Teece et al., 1997).

These theories keep the proposal from treating AI as a fashionable label. They show that digital transformation creates value only when it strengthens strategic capabilities and supports coherent managerial action. Their limitation is just as important. Classical strategic-management frameworks were not designed to assess AI governance, data-risk exposure, ethical legitimacy, or smart-city stakeholder complexity. They explain why capabilities matter, but they do not give managers a measurable readiness model for deciding whether an AI-enabled initiative is strategically, financially, operationally, and ethically ready for implementation.

5.2 Digital Transformation as Business and Organizational Change

Digital transformation literature moves the argument from strategy to technology-enabled organizational change. Bharadwaj et al. argue that digital business strategy fuses IT strategy with business strategy, so technology can no longer be treated as a separate support function (Bharadwaj et al., 2013). Vial frames digital transformation as a process in which digital technologies disrupt organizations and force changes in value creation and structure (Vial, 2019). Verhoef et al. distinguish digitization, digitalization, and full digital transformation, showing that real transformation changes business models, capabilities, and performance metrics, not merely technical processes (Verhoef et al., 2021).

This stream matters because it explains why AI-enabled transformation cannot be judged through technical adoption alone. An initiative may deploy advanced AI tools and still fail as a business transformation if it lacks investment logic, governance discipline, organizational readiness, or stakeholder legitimacy. The weakness of much digital-transformation literature is its breadth. It identifies drivers, capabilities, and stages of transformation, but rarely turns them into an DTET-level scoring model that can compare concrete investment scenarios under defined constraints.

5.3 AI Governance and the Problem of Managerial Control

AI governance literature adds the control layer that managerial decision-making needs. NIST and OECD-style frameworks emphasize accountability, transparency, robustness, human oversight, risk management, and trustworthy deployment (National Institute of Standards and Technology, 2023; Organisation for Economic Co-operation and Development, n.d.). Related AI ethics literature shows convergence around fairness, privacy, accountability, transparency, and human-centered values (Floridi et al., 2018; Jobin et al., 2019).

This literature strengthens the proposal because AI-enabled transformation creates risks that traditional strategy tools do not fully capture: opaque decision-making, data misuse, automation bias, accountability gaps, weak documentation, and stakeholder distrust. Yet AI-governance frameworks also leave a practical measurement problem. How should a digital-transformation researcher score the readiness of an AI-enabled initiative when the same initiative must satisfy strategy, ROI, operations, governance, and ethical criteria at once? This unresolved measurement problem justifies the integrated framework.

5.4 Performance Measurement and ROI Discipline

Performance-measurement literature brings the financial and managerial discipline required for this study. Kaplan and Norton's balanced scorecard broadened performance evaluation beyond financial results by integrating customer, internal-process, and learning dimensions (Kaplan & Norton, 1992). That logic matters for AI-enabled transformation because a narrow ROI calculation can miss strategic learning, risk reduction, operational resilience, and stakeholder value. AHP-style multi-criteria reasoning also supports the need to compare heterogeneous criteria instead of reduce readiness to one financial indicator (Saaty, 1980).

Performance-measurement tools still have limits in this context. The balanced scorecard was not designed for AI-enabled transformation, smart-city governance, Saudi regulatory expectations, or Islamic business-ethics criteria. Traditional ROI tools can measure financial discipline, but they do not automatically capture data governance, ethical legitimacy, or public-value alignment. This proposal therefore treats ROI discipline as one dimension of readiness, not as the whole evaluation model.

5.5 Saudi Smart-City Context and Institutional Requirements

The Saudi context changes what digital transformation readiness means. Vision 2030, SDAIA, PDPL materials, Study in Saudi, Prince Sattam bin Abdulaziz University's digital-transformation program context, and NEOM-related public materials position digital transformation as part of a national modernization, competitiveness, and governance agenda (Ministry of Education, Kingdom of Saudi Arabia, n.d.; Prince Sattam bin Abdulaziz University, n.d.; NEOM, n.d.; Saudi Data and Artificial Intelligence Authority, n.d.-a; Saudi Data and Artificial Intelligence Authority, n.d.-c; Saudi Vision 2030, n.d.). In this setting, AI-enabled initiatives cannot be assessed only as private-sector efficiency projects. They also need to satisfy institutional alignment, data-protection obligations, stakeholder trust, and public-value expectations.

That contextual layer gives the proposal its originality. Generic digital-transformation models can identify technological and organizational readiness, but they do not fully capture Saudi institutional conditions. Official Saudi policy sources define ambition and regulatory direction, but they do not provide an digital-transformation research instrument for comparing AI-enabled business decision

scenarios. The resulting methodological gap is precise: the literature lacks a bounded, measurable, Saudi-contextual readiness framework that translates policy, strategy, governance, and performance criteria into a practical assessment model.

5.6 Islamic Business Ethics as Operational Criteria

Islamic business ethics adds a normative layer that matters in a Saudi digital-transformation and emerging-technologies proposal. Concepts such as Al-Amanah, Ihsan, and Maslaha can strengthen the evaluation of AI-enabled transformation by foregrounding trustworthy stewardship, excellence in execution, and public interest. Raquib et al. show that Islamic ethical reasoning can contribute to AI ethics by connecting technological responsibility to moral accountability (Raquib et al., 2022).

The difficulty is operational. Islamic ethics is often discussed conceptually, while digital-transformation research requires observable criteria. Al-Amanah cannot remain a decorative ethical label; it must become auditable through responsibility assignment, truthful documentation, data stewardship, and accountability trails. Ihsan must be measurable through quality monitoring, continuous improvement, and error-response mechanisms. Maslaha must be linked to stakeholder value, harm reduction, accessibility, and public-interest justification. This research therefore does not merely mention Islamic ethics; it tests whether these principles add analytical coverage beyond generic governance indicators.

5.7 Literature Synthesis Matrix

Table 3 synthesizes the literature thematically and shows that the research gap lies at the intersection of strategy, digital transformation, AI governance, performance measurement, Saudi smart-city context, and Islamic business ethics.

Table 3. Thematic Literature Synthesis and Identification of the Research Gap

Research stream	Core contribution	Critical limitation	Implication for this study
Strategic management (Barney, 1991; Porter, 1985; Teece et al., 1997)	Explains value creation, competitive advantage, resources, and dynamic capabilities.	Does not operationalize AI governance, data-risk exposure, stakeholder trust, or Saudi smart-city conditions.	Provides the strategic-alignment and capability logic for the readiness framework.
Digital transformation (Bharadwaj et al., 2013; Verhoef et al., 2021; Vial, 2019)	Shows that digital technologies reshape business models, capabilities, processes, and performance logic.	Often remains broad and does not provide scenario-based managerial scoring for AI-enabled initiatives.	Supports the transformation-readiness dimensions and distinguishes true transformation from simple technology adoption.
AI governance and AI ethics (Floridi et al., 2018; Jobin et al., 2019; National Institute of Standards and Technology, 2023; Organisation for Economic Co-operation and Development, n.d.)	Defines accountability, transparency, risk management, robustness, privacy, and human oversight principles.	Principles are often stated at a high level and are not integrated with ROI discipline or DTET performance assessment.	Provides governance and risk-control criteria for the scoring rubric.
Performance measurement and decision tools (Kaplan & Norton, 1992;	Broadens evaluation beyond narrow financial results and supports multi-criteria assessment.	Traditional tools do not fully capture AI-specific governance, Saudi regulatory context, or Islamic ethics.	Supplies ROI discipline, non-financial indicators, and multi-criteria scoring logic.

Saaty, 1980)			
Saudi institutional and smart-city context (Ministry of Education, Kingdom of Saudi Arabia, n.d.; Prince Sattam bin Abdulaziz University, n.d.; NEOM, n.d.; Saudi Data and Artificial Intelligence Authority, n.d.-a; Saudi Data and Artificial Intelligence Authority, n.d.-c; Saudi Vision 2030, n.d.)	Frames digital transformation through Vision 2030, data governance, education/program fit, and NEOM as a bounded case setting.	Official sources state ambition and policy direction but do not offer a research-tested DTET assessment instrument.	Justifies the Saudi-contextual boundaries and the NEOM-based synthetic scenario design.
Islamic business ethics (Raquib et al., 2022)	Adds stewardship, excellence, public interest, and moral accountability to business and AI evaluation.	Often remains normative unless translated into observable audit criteria.	Creates a distinct ethical coverage layer through Al-Amanah, Ihsan, and Maslaha audit items.

Note. The table synthesizes selected foundational and recent sources by theme. It is not a chronological author list; it identifies contribution, limitation, and implication for the proposed framework.

5.8 Synthesis and Research Gap

The reviewed literature is strong within separate domains but weak where those domains meet. Strategic management explains value creation and capabilities, but it does not operationalize AI governance. Digital-transformation literature explains organizational change, but it often remains too broad for scenario-based managerial scoring. AI-governance literature identifies accountability and risk principles, but it rarely integrates ROI discipline and business-performance readiness. Performance-measurement literature offers financial and non-financial evaluation logic, but it does not fully address AI-specific governance or Saudi smart-city conditions. Islamic business ethics contributes culturally grounded legitimacy, but it often lacks measurable audit criteria.

The research gap is therefore not simply the absence of another framework. The gap is the lack of evidence on whether an integrated digital-transformation-oriented readiness framework can assess AI-enabled digital transformation initiatives more comprehensively than a generic digital-transformation checklist. This research addresses that gap by translating fragmented literature into measurable variables, scoring criteria, Islamic-ethics audit items, and synthetic NEOM-based business decision scenarios. The literature leads directly to the main research question: whether an integrated framework improves the assessment of strategic performance readiness under bounded Saudi smart-city conditions.

6. THEORETICAL AND CONCEPTUAL FRAMEWORK

6.1 Theoretical Position of the Study

This project is best understood as a theory-refinement and operationalization study. It is not a pure theory-testing project, and it is not grounded theory. It does not try to build a new theory of digital transformation from scratch. Instead, it uses established theories from strategic management, digital

transformation, performance measurement, AI governance, and Islamic business ethics to build and evaluate an integrated digital-transformation-oriented readiness framework.

The primary theoretical lens is Dynamic Capabilities Theory. Teece, Pisano, and Shuen define dynamic capabilities as the firm's ability to integrate, build, and reconfigure internal and external competences in changing environments (Teece et al., 1997). This lens is appropriate because AI-enabled digital transformation requires organizations to sense digital opportunities, seize them through disciplined investment and execution, and reconfigure processes, governance structures, and stakeholder relationships. Teece's later microfoundations work further supports this logic by connecting sensing, seizing, and reconfiguration to sustainable enterprise performance (Teece, 2007).

The Resource-Based View supports this primary lens. Barney argues that resources create sustained advantage only when they are valuable, rare, imperfectly imitable, and non-substitutable (Barney, 1991). In this study, AI is therefore not treated as valuable by itself. It becomes managerially meaningful only when it is connected to strategic capabilities, operational execution, governance discipline, and value creation.

Digital transformation theory defines the phenomenon being assessed. Vial conceptualizes digital transformation as a process in which digital technologies create disruptions that trigger strategic responses and changes in value creation paths (Vial, 2019). Verhoef et al. distinguish digitization, digitalization, and full digital transformation, which supports the study's decision to evaluate transformation as a business-model and capability problem instead of a simple technology-adoption issue (Verhoef et al., 2021).

Performance-measurement theory provides the evaluation logic. Kaplan and Norton argue that managerial performance assessment should combine financial and non-financial indicators instead of rely on a single accounting measure (Kaplan & Norton, 1992). This justifies a multidimensional readiness model that includes investment discipline, operational feasibility, governance control, stakeholder trust, and ethical legitimacy.

AI governance provides the control and accountability layer. NIST AI RMF 1.0 structures AI risk management around Govern, Map, Measure, and Manage functions (National Institute of Standards and Technology, 2023). OECD AI principles similarly emphasize trustworthy and human-centred AI that respects rights, transparency, robustness, and accountability (Organisation for Economic Co-operation and Development, n.d.). These sources justify the inclusion of documentation, oversight, traceability, accountability, and risk-control criteria in the framework.

Islamic business ethics provides the contextual legitimacy layer. Islamic virtue-based AI ethics argues that AI governance should not rely only on technical efficiency or market logic, but should also address moral responsibility, stewardship, and public-interest concerns (Raquib et al., 2022). In this study, Al-Amanah, Ihsan, and Maslaha are operationalized as auditable managerial criteria: trustworthy stewardship, excellence in execution, and public-interest value.

6.2 How the Theories Explain the Research Problem

The research problem exists because existing assessment approaches view AI-enabled transformation through fragmented lenses. Strategic management explains value creation but does not fully capture AI governance. Digital transformation literature explains organizational change but often lacks detailed risk and ethics scoring. Performance-measurement frameworks broaden evaluation beyond ROI but do not directly operationalize AI-specific accountability. AI governance frameworks define trustworthiness and risk control but do not normally integrate DTET-oriented financial and strategic readiness. Islamic ethics adds moral and contextual legitimacy, but it often remains normative unless translated into measurable criteria.

This theoretical fragmentation explains why the project is necessary. The central proposition is modest and testable: a framework that integrates these lenses should provide stronger strategic performance-readiness assessment quality than a generic digital-transformation checklist. The claim is not that the framework improves real project performance. The narrower claim is that it can improve the analytical coverage of managerial assessment under controlled synthetic scenario conditions.

6.3 Theoretical Mapping to Variables

Table 4 translates the theoretical framework into measurable dimensions, ensuring that no variable is disconnected from the study's theoretical foundations.

Table 4. *Theoretical Foundations, Derived Variables, and Assessment Dimensions*

Theoretical foundation	Core idea	Variable or dimension derived	Role in the study
Dynamic Capabilities Theory	Organizations must sense, seize, and reconfigure capabilities in changing environments.	Strategic alignment; operational feasibility; transformation readiness.	Primary theoretical spine.
Resource-Based View	Resources create value only when they become strategic capabilities.	Capability fit; resource-value logic; investment discipline.	Explains why AI is not automatically valuable.
Digital Transformation Theory	Digital technologies reshape value creation, business models, and organizational structures.	AI-enabled digital transformation readiness.	Defines the phenomenon being assessed.
Balanced Scorecard / Performance Measurement	Performance assessment must combine financial and non-financial dimensions.	Strategic performance-readiness assessment quality.	Justifies multidimensional scoring.
AI Governance Frameworks	Trustworthy AI requires governance, risk control, accountability, transparency, and oversight.	Governance-risk control; stakeholder trust; auditability.	Justifies control and accountability dimensions.
Islamic Business Ethics	Business and technology decisions should reflect stewardship, excellence, and public interest.	Islamic ethical legitimacy: Al-Amanah, Ihsan, Maslaha.	Adds contextual and ethical audit criteria.

Note. RBV = resource-based view; AI = artificial intelligence; DTET = Digital Transformation and Emerging Technologies. Each theoretical lens is linked to at least one assessment dimension to avoid unsupported variables.

6.4 Conceptual Logic of the Model

The figures in this proposal are designed as methodological and conceptual aids, not decorative illustrations. Each figure clarifies the research logic, variable structure, workflow sequence, or timeline-control mechanism; none of the current figures reports empirical statistical results.

The model postulates that the type and integration level of the assessment framework influence the quality of strategic performance-readiness assessment. The independent variable is assessment framework type / integration level. It compares a generic digital-transformation checklist with the proposed integrated digital-transformation-oriented readiness framework.

The dependent variable is strategic performance-readiness assessment quality. It refers to the extent to which the assessment captures the dimensions required for managerial decision-making: strategy, investment logic, operational feasibility, governance-risk control, stakeholder trust, and Islamic ethical legitimacy.

The assessment dimensions function as the explanatory bridge between the independent and dependent variables. The integrated framework is expected to provide stronger analytical coverage because it combines theoretical lenses that generic checklists usually treat separately.

Figure 1 clarifies the central logic of the study: framework type is expected to influence assessment quality through six theoretically grounded assessment dimensions.

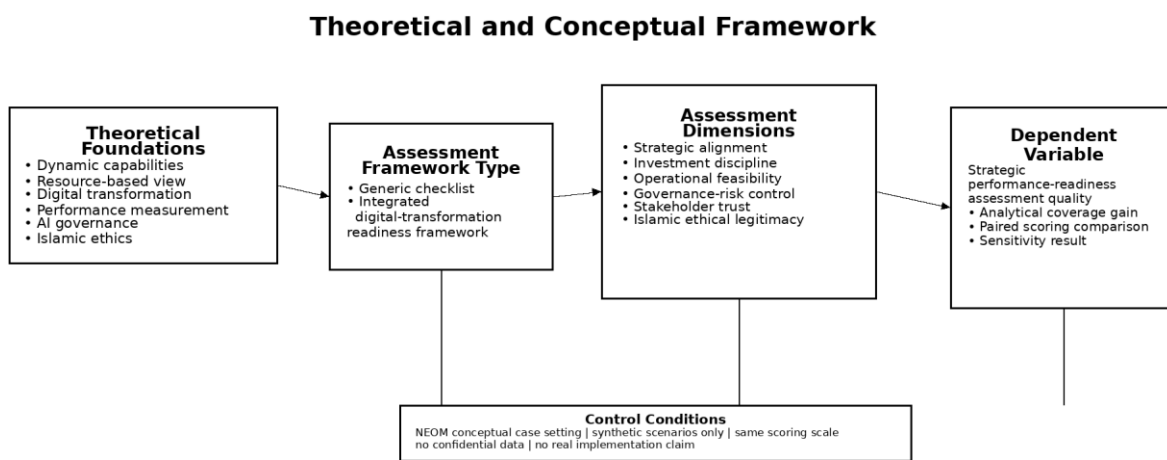


Figure logic: theories define assessment dimensions; framework integration changes assessment coverage, not real NEOM performance.

Figure 1. *Theoretical and Conceptual Framework Linking Assessment Integration Level to Strategic Performance-Readiness Assessment Quality*

Note. AI = artificial intelligence; DTET = Digital Transformation and Emerging Technologies; NEOM is used as a bounded conceptual case setting, not as an object of operational audit. The figure shows how theoretical foundations inform the assessment framework type, which is translated into six assessment dimensions: strategic alignment, investment discipline, operational feasibility, governance-risk control, stakeholder trust, and Islamic ethical legitimacy. The dependent variable is strategic performance-readiness assessment quality. The figure does not claim to measure real NEOM performance, financial outcomes, citizen adoption, or technical AI accuracy.

6.5 Refutable Theoretical Propositions

Table 5 converts the conceptual model into refutable propositions and corresponding test logic.

Table 5. *Refutable Theoretical Propositions, Null Propositions, and Test Logic*

Proposition	Null proposition	Test logic
P1. The integrated digital-transformation-oriented readiness framework will provide broader analytical coverage of strategic performance readiness than a generic digital-transformation checklist.	H0-1. There is no difference in analytical coverage between the integrated framework and the generic checklist.	Paired comparison across the same synthetic NEOM-based business decision scenarios.
P2. The expected coverage advantage will result from the integration of strategic, financial, operational, governance, stakeholder-trust, and Islamic-ethics dimensions.	H0-2. These additional dimensions do not explain any observable difference in assessment coverage.	Dimension-level score decomposition and coverage-gain analysis.
P3. Islamic business-ethics criteria will add	H0-3. Islamic business-ethics criteria do not	Coverage-overlap matrix comparing generic

distinguishable audit coverage beyond generic governance indicators when translated into observable criteria.	add distinguishable analytical coverage beyond generic governance criteria.	governance indicators with Al-Amanah, Ihsan, and Maslaha audit items.
P4. Synthetic NEOM-based business decision scenarios can support preliminary internal validation when confidential operational data are unavailable.	H0-4. Synthetic scenarios cannot provide repeatable preliminary validation.	Fixed scenario assumptions, identical scoring procedure, and reproducibility checks.

Note. The propositions concern analytical coverage under synthetic scenario conditions. They do not claim to test real NEOM implementation performance or causal project outcomes.

6.6 Justification Against Alternative Theories

Alternative theories were considered but are not used as the primary lens. Technology-adoption theories are less appropriate because this study does not examine whether individual users accept AI systems. Agency theory is relevant to accountability but too narrow to explain strategic transformation readiness. Institutional theory is useful for explaining conformity to Saudi policy and regulation, but it does not sufficiently capture managerial capability, investment discipline, and performance assessment. Therefore, Dynamic Capabilities Theory, supported by RBV, digital transformation theory, performance measurement, AI governance, and Islamic ethics, provides the strongest theoretical architecture for this proposal.

6.7 Boundary Conditions

The theoretical framework is deliberately bounded. It does not explain actual NEOM performance, macroeconomic impact, real citizen adoption, technical AI accuracy, or long-term institutional outcomes. Those phenomena would require confidential data, longitudinal measurement, and external validation. This research evaluates only whether an integrated framework improves assessment coverage under synthetic, controlled, and reproducible digital-transformation research conditions.

7. RESEARCH METHODOLOGY

7.1 Research Design and Methodological Position

The study adopts a pragmatic mixed-methods, framework-building, and comparative case-scenario design. The design is pragmatic because the research question requires both interpretation and structured measurement: first, the project identifies strategic, governance, financial, operational, stakeholder-trust, and Islamic-ethics criteria from documents and literature; then it tests whether those criteria improve assessment coverage when applied to the same synthetic business decision scenarios. Mixed-methods research is appropriate because the qualitative and quantitative strands answer different parts of the question, and neither strand is sufficient alone (Creswell & Creswell, 2018).

The case logic is bounded instead of operational. NEOM serves as a conceptual Saudi smart-city megaproject case setting, not as a field site for auditing implementation. This follows case-study logic: the case must be defined, the evidence sources must be bounded, and the design must protect validity through a clear chain of evidence (Yin, 2018). The project therefore evaluates assessment quality under controlled scenario conditions. It does not measure real NEOM performance, citizen adoption, procurement outcomes, or confidential institutional decisions.

7.2 Methodology Versus Methods

Methodology refers to the overall research logic: a controlled framework-building design supported by document analysis, theory-based operationalization, synthetic scenario construction, paired scoring, and robustness checks. Methods are the instruments inside that logic: a source log, literature matrix, codebook, 0-3 scoring rubric, Islamic-ethics mapping sheet, synthetic scenario template, paired comparison workbook, and sensitivity-analysis table. This distinction keeps the design from becoming a loose list of tools.

7.3 Unit of Analysis, Population, and Sampling Frame

The primary unit of analysis is not a person, organization, or real NEOM project. It is the assessment of a synthetic AI-enabled digital transformation decision scenario under two alternative tools: a generic digital-transformation checklist and the proposed integrated digital-transformation-oriented readiness framework. The comparison therefore occurs at the scenario-assessment level.

The sampling strategy has three layers. First, the project will use purposive theoretical sampling of at least 15-20 official, academic, and digital-transformation and emerging-technologies sources that directly address digital transformation, strategic management, AI governance, performance measurement, Saudi smart-city context, and Islamic business ethics. Second, it will construct at least 30 synthetic NEOM-based business decision scenarios. Third, optional expert or supervisor review may be used only for face-validity checking after ethics clearance; if approved, the target will be 3-5 reviewers with relevant expertise in digital transformation and emerging technologies, AI governance, digital transformation, or Saudi policy.

Table 6 makes the study reproducible by specifying which sources and scenario units are eligible for analysis and which sources are excluded.

Table 6. *Units of Analysis, Sampling Frame, Inclusion Criteria, and Research Instruments*

Research component	Sampling / unit of analysis	Instrument	Output
RQ1: Identify readiness requirements	15-20 purposively selected official, academic, and digital-transformation and emerging-technologies sources	Source log; literature matrix; inclusion/exclusion checklist	Coded source matrix and requirement categories
RQ2: Operationalize variables and scoring	Coded requirements from RQ1; theoretical constructs from Section 6	Codebook; variable table; 0-3 scoring rubric	Operational variables and scoring criteria
RQ3: Map Islamic ethics criteria	Al-Amanah, Ihsan, and Maslaha as ethical constructs; governance checklist criteria as comparator	Ethics mapping sheet; overlap matrix	Auditable Islamic business-ethics criteria and coverage notes
RQ4: Evaluate framework coverage	At least 30 synthetic NEOM-based business decision scenarios scored under both tools	Scenario template; paired scoring workbook; optional Python/Excel file	Paired scoring table, coverage-gain matrix, and sensitivity results

Note. The project uses public documents, academic literature, coded criteria, and synthetic business decision scenarios. Optional human input is subject to institutional ethics approval, exemption, or non-human-subjects determination.

7.4 Data Collection Instruments

The project collects evidence through structured, non-confidential instruments. The source log records citation details, access date, source type, relevance, and exclusion reason where applicable. The literature matrix extracts theory, constructs, methods, variables, and limitations from each

source. The codebook defines each assessment dimension: strategic alignment, investment discipline, operational feasibility, governance-risk control, stakeholder trust, and Islamic ethical legitimacy. The 0-3 rubric turns these dimensions into observable scoring criteria. The synthetic scenario template specifies assumptions, decision context, investment logic, operational complexity, data-governance risk, stakeholder implications, and ethics-related considerations.

No confidential NEOM data, personal mobility data, proprietary AI model outputs, or internal financial records will be collected. If optional human review is later added, no interview, survey, review form, or identifiable expert input will be collected before ethics approval or formal exemption has been documented.

7.5 Pilot Procedure and Scenario Construction

Before full scoring, the rubric will be pilot-tested on 3-5 synthetic scenarios. The pilot will check whether each criterion is interpretable, non-duplicative, and scoreable. Criteria that create ambiguity, double-counting, or unjustified score inflation will be revised before the full scenario set is locked. After the pilot, at least 30 synthetic scenarios will be generated using fixed assumptions and documented variation across strategic alignment, investment logic, operational feasibility, data-governance exposure, stakeholder trust, and Islamic ethical legitimacy. The scenario set will then be frozen before comparative scoring so that cases cannot be adapted to favor the proposed framework.

7.6 Data Analysis Plan

The analysis proceeds in six steps. First, directed content analysis identifies requirements from official and academic sources. Second, thematic synthesis groups those requirements into assessment dimensions. Third, variable operationalization converts dimensions into scored criteria. Fourth, paired scenario scoring applies the generic checklist and the integrated DTET framework to the same scenarios. Fifth, coverage analysis compares overlapping and non-overlapping criteria to determine whether additional dimensions create distinct analytical value. Sixth, sensitivity analysis tests whether the findings remain stable under alternative weights, normalized scores, or removal of overlapping criteria. AI governance controls will be benchmarked against NIST AI RMF logic and OECD trustworthy-AI principles where relevant (National Institute of Standards and Technology, 2023; Organisation for Economic Co-operation and Development, n.d.).

Table 7 links each data type to its analytical method, expected output, and bias-control mechanism.

Table 7. Data Analysis Strategy, Outputs, and Bias-Control Procedures

Data / evidence type	Analysis method	Metric or output	Validity / bias control
Official and academic sources	Directed content analysis and thematic synthesis	Requirement map; literature-policy matrix	Predefined inclusion criteria; source log; no uncited claims
Strategic and performance constructs	Theory-based operationalization	Variable table and 0-3 scoring rubric	Trace each criterion to a theory, source, or research question
Islamic business-ethics criteria	Concept-to-control mapping and overlap comparison	AI-Amanah, Ihsan, Maslaha audit criteria; coverage-overlap matrix	Separate cultural framing from distinct measurable audit value
Synthetic business scenarios	Pilot testing followed by paired scoring under two assessment tools	Scenario scores; raw and normalized comparison	Same scenarios, same scale, frozen assumptions, versioned scoring file
Comparative results	Coverage-gain analysis and sensitivity testing	Coverage-gain matrix; sensitivity table; negative-result interpretation	Distinguish score inflation from genuine analytical coverage

Note. The table distinguishes qualitative coding, rubric-based scoring, paired comparison, coverage analysis, and sensitivity testing. The planned outputs are audit tools, not operational NEOM measurements.

7.7 Sample Size and Feasibility Logic

A target of at least 30 synthetic decision scenarios is appropriate for an exploratory DTET framework-evaluation study because the objective is not to estimate real-world NEOM effects or produce population-level statistical inference. The objective is to test whether the same scenario set is assessed more comprehensively by the integrated framework than by a generic checklist. If time permits, an optional simulated project-record dataset of 500-1,000 synthetic rows may be used for illustrative descriptive analysis, but this component remains non-essential and will not replace the paired scenario comparison.

7.8 Reproducibility and Quality Controls

Reproducibility will be protected through a fixed source log, versioned codebook, documented scenario assumptions, identical scenario sets for both frameworks, consistent scoring scales, frozen pilot revisions, and a transparent audit trail. Scoring files will record date, scorer, scenario identifier, criterion, raw score, normalized score, and justification note. The project will also maintain a negative-result protocol: if the integrated framework does not outperform the generic checklist after normalization, the finding will be reported as boundary-condition evidence instead of suppressed.

7.9 Rejected Alternative Methods and Methodological Justification

A full field study inside NEOM was rejected because it would require confidential institutional access and would shift the thesis from framework evaluation to operational audit. A purely qualitative study was rejected because it would not test the comparative assessment value of the framework. A purely quantitative or financial model was rejected because it would ignore governance, stakeholder trust, and Islamic ethical legitimacy. A user-adoption survey was rejected because the project does not investigate individual acceptance of AI systems. The selected design is therefore the strongest feasible design for the research question: it combines theoretical synthesis, measurable operationalization, and controlled comparative evaluation without overclaiming real-world implementation effects.

7.10 Methodological Workflow

Figure 2 presents the methodological workflow as a sequence of auditable research operations, showing how each stage produces an output required by the next stage.

Methodological Workflow

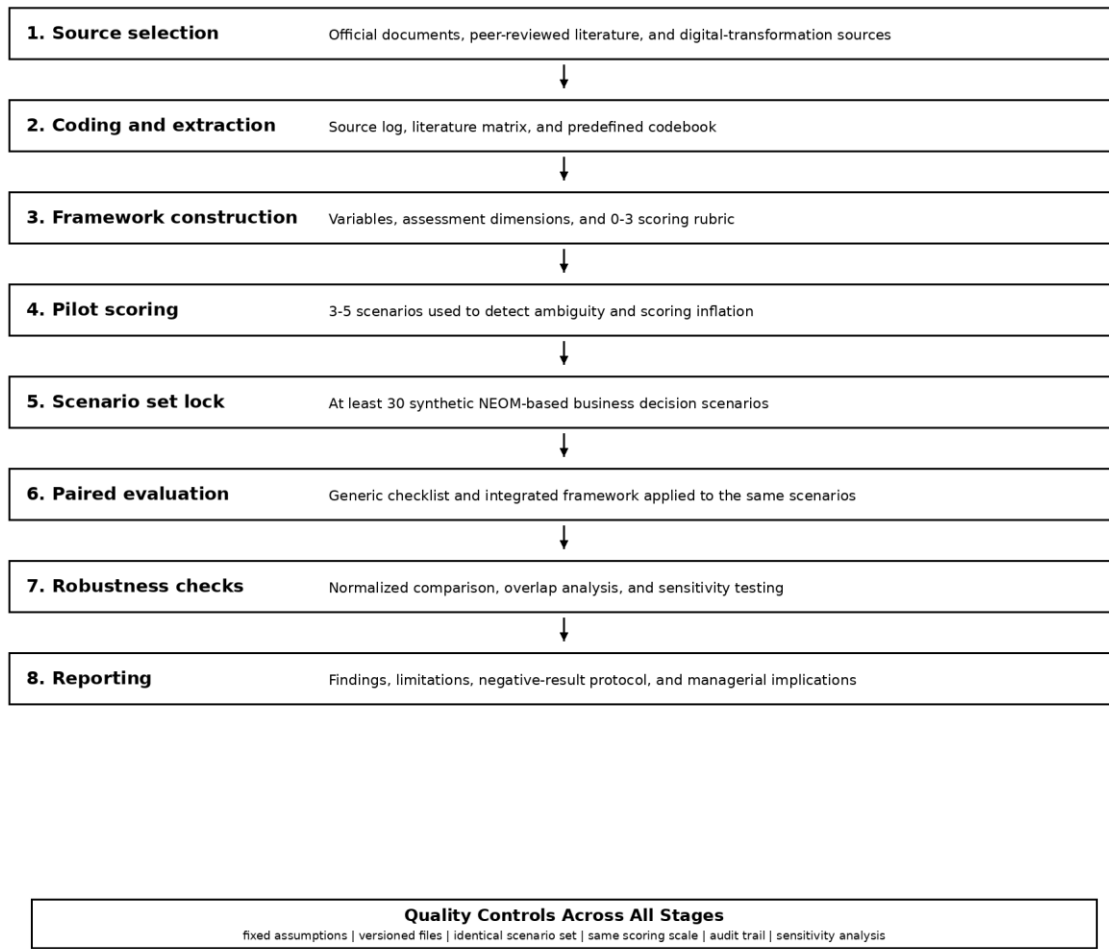


Figure 2. Methodological Workflow for Developing and Evaluating the Integrated DTET Readiness Framework

Note. DTET = Digital Transformation and Emerging Technologies. The workflow follows a sequential logic: source selection, literature and policy coding, codebook construction, framework development, pilot scoring, synthetic scenario construction, paired scoring against the generic checklist, sensitivity analysis, interpretation, and reporting. Solid arrows indicate the main analytical sequence. Feedback loops indicate revision points after pilot testing and robustness checks. No confidential NEOM data, personal data, or real operational datasets are used.

The detailed source selection protocol, literature review matrix, coding protocol, scoring rubric, ethics mapping, synthetic scenario protocol, scenario bank, baseline checklist, paired scoring sheet, and sensitivity-analysis template are provided in Appendices A-J to support methodological transparency and reproducibility.

8. VALIDATION, VERIFICATION, AND EVALUATION RUBRIC

8.1 Validation Mechanism

The expected outcome will be evaluated through three checks: document benchmarking against official and academic sources, synthetic scenario comparison against a generic baseline checklist, and reproducibility verification through documented assumptions, fixed scenario templates, and versioned scoring sheets. The design tests the null hypotheses instead of only seeking confirmatory evidence.

8.2 Scenario Firewall and Benchmarking

Literature and policy sources generate the framework. Synthetic decision scenarios that did not construct the framework evaluate it. When needed, the variants are split into development and evaluation groups. The baseline is a generic digital-transformation checklist derived from broad transformation and governance principles without the Saudi context, business-case logic, and Islamic ethics layers. The study compares the proposed integrated framework against this baseline.

8.3 Reviewer-Simulation Rubric

Table 8 provides a reviewer-simulation rubric that functions as a proposal quality-control tool instead of as a decorative evaluation table.

Table 8. Reviewer-Simulation Evaluation Rubric for Framework Quality Control

Criterion	0 = fail	1 = weak	2 = adequate	3 = strong
GIFT - Gap	No clear gap.	Broad gap only.	Specific gap stated.	Quantified, delimited, literature-backed gap.
GIFT - Idea	Idea vague.	Idea interesting but untestable.	Idea is feasible.	Idea is original, bounded, and evaluable.
GIFT - Feasibility	Methods absent.	Methods listed without justification.	Methods justified.	Methods are reproducible, resourced, and risk-managed.
GIFT - Timeline	No timeline.	Unrealistic timeline.	12-month plan.	Milestone-based plan linked to deliverables.
program fit	Purely technical or unrelated.	Weak business link.	Business value stated.	Strategy, finance, operations, governance, and stakeholder value are integrated.
Reproducibility	Cannot replicate.	Partial detail.	Replicable with effort.	Another researcher can reproduce workflow.
ROI and performance	No value logic.	Value unclear.	Useful outputs.	Clear financial, operational, and strategic value logic.
Fatal flaws	Overclaiming or incoherent.	Major unresolved mismatch.	Minor limitations stated.	Limitations controlled and no overclaiming.

Note. The rubric is a reviewer-simulation quality-control tool for proposal readiness. It is not the same as the final empirical scoring rubric used for synthetic scenario comparison.

8.4 Risk Management and Contingency Plan

This research uses a controlled risk-management approach because the project depends on public documents, coded literature, synthetic business decision scenarios, and comparative scoring. The plan separates minor execution issues from critical methodological threats. Risks are classified by probability and impact, then linked to mitigation and contingency actions. This follows standard project-risk practice, where probability-impact assessment, mitigation planning, contingency responses, and tracking protect delivery (Project Management Institute, 2019).

The central risk-management principle is methodological continuity. If the primary design cannot be executed exactly as planned, the project will shift to a documented fallback design while preserving the core research question: whether an integrated digital-transformation-oriented readiness framework provides stronger analytical coverage than a generic digital-transformation checklist under bounded Saudi smart-city conditions.

Table 8A identifies the main threats to validity and shows how each threat will be mitigated before it affects the credibility of the findings.

Table 8A. Risk Register, Contingency Controls, and Residual Risk Position

Risk identified	Probability	Impact	Trigger indicator	Mitigation and contingency plan
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Risk identified	Probability	Impact	Trigger indicator	Mitigation and contingency plan
R1. Public-document limitation	Medium	Medium	Official sources are promotional, incomplete, or too general for robust coding.	Triangulate official sources with peer-reviewed literature on digital transformation, AI governance, performance measurement, and smart-city management. If official evidence is weak, treat it as contextual policy narrative rather than empirical proof.
R2. No access to confidential NEOM or institutional data	High	Low-Medium	No permission is available for internal documents, real project metrics, or stakeholder interviews.	Design the study from the start as a public-document and synthetic-scenario project. NEOM remains a bounded conceptual case setting. Findings will be reported as preliminary internal validation, not operational validation.
R3. Synthetic scenarios challenged as unrealistic	Medium	High	Reviewers argue that synthetic scenarios cannot represent megaproject complexity.	Document assumptions, scenario variables, scoring rules, and boundary conditions. Run sensitivity checks. If challenged, reframe findings as scenario-based evidence requiring future expert-panel or institutional validation.
R4. Criterion inflation / scoring bias	Medium	High	The integrated framework appears better only because it contains more criteria than the generic checklist.	Normalize scoring scales, use the same scenarios for both tools, distinguish score level from coverage gain, and compare overlapping versus non-overlapping criteria. If the advantage disappears, report framework complexity as a limitation.
R5. Weak distinction between Islamic ethics and generic governance	Medium	High	Al-Amanah, Ihsan, and Maslaha appear to duplicate accountability, quality management, or stakeholder value.	Translate each principle into observable audit items and compare them against generic governance indicators. If overlap is high, report that Islamic ethics contextualizes rather than independently expands the framework.
R6. Coding subjectivity	High	Medium	Source coding, ethics mapping, or scenario scoring depends too strongly on researcher judgment.	Use a predefined codebook, audit trail, versioned scoring rubric, and source log. Request supervisor or independent review of a sample if available. If not available, disclose the single-coder limitation.
R7. Statistical comparison is underpowered or inappropriate	Medium	Medium	The scenario sample is too small or test assumptions are not met.	Use at least 30 synthetic scenarios where feasible. If inferential testing is inappropriate, replace it with descriptive paired scoring, coverage matrices, and sensitivity tables, and label the result exploratory.
R8. Timeline compression	Medium	Medium	Literature coding, framework design, or scenario scoring takes longer than planned.	Lock the codebook by Month 5 and avoid adding new objectives. If time compresses, reduce the scenario set to a justified pilot while preserving the same comparative logic.

Risk identified	Probability	Impact	Trigger indicator	Mitigation and contingency plan
R9. AI governance or Saudi data-policy update during the project	Low-Medium	Medium	New official guidance or regulatory updates appear after framework construction.	Maintain a dated source log. NIST AI RMF and OECD principles remain stable benchmark frameworks for AI risk and trustworthy AI comparison (National Institute of Standards and Technology, 2023; Organisation for Economic Co-operation and Development, n.d.). If an update is material, add a regulatory update note rather than redesigning the full study.
R10. Negative or mixed results	Medium	Medium	The integrated framework does not outperform the generic checklist across all dimensions.	Predefine the interpretation logic. The study evaluates analytical coverage, not guaranteed superiority. Negative or mixed results will be reported as boundary-condition evidence and used to refine the framework.

Note. The most critical methodological risk is criterion inflation, where the integrated framework may appear stronger simply because it contains more assessment criteria than the generic checklist.

Worst-case scenario protocol

The worst credible scenario is that the integrated framework produces higher raw scores than the generic checklist, but the advantage disappears after normalization. This would not destroy the thesis. It would shift the contribution from framework superiority to boundary-condition evidence. The dissertation would report that additional DTET, governance, and Islamic-ethics dimensions did not produce sufficient distinguishable coverage under synthetic scenario conditions.

That negative result would still be useful because it would show where integrated readiness frameworks become redundant, over-complex, or vulnerable to criterion inflation. The project would then refine the framework by separating genuinely new analytical coverage from criteria that merely rename existing governance or performance concepts.

Residual-risk position

After mitigation, the project remains feasible because it does not depend on confidential NEOM data, real-time system access, large-scale human recruitment, expensive software, or external funding. The core deliverables can be completed through public sources, peer-reviewed literature, a transparent codebook, synthetic business decision scenarios, paired scoring, and sensitivity analysis. The strongest residual risk is not data access; it is proving that added criteria create measurable analytical coverage instead of cosmetic complexity.

9. ETHICAL CONSIDERATIONS AND DATA PROTECTION

This section defines the ethical, legal, and data-protection controls for the research. The core evidence is low-risk: public documents, peer-reviewed literature, coded criteria, and synthetic NEOM-based business decision scenarios. The project does not require confidential NEOM data, real personal mobility traces, patient records, minors, vulnerable populations, or commercially sensitive institutional datasets. If optional expert review is later added, no recruitment or data collection will begin before formal university ethics approval, exemption, or non-human-subjects determination has been documented.

9.1 Institutional ethics approval

The researcher will submit the complete protocol to the relevant Research Ethics Committee, Institutional Review Board, or university-approved ethics authority before collecting any data from human participants. The submission package will include the research purpose, sampling criteria, consent form, participant information sheet, interview or review instrument if applicable, data-management plan, confidentiality procedures, risk-benefit assessment, and withdrawal procedure. No interview, expert review, survey, recording, or identifiable data collection will occur before formal approval is granted.

9.2 Data scope and participant involvement

The primary study uses three low-risk data categories: publicly available institutional documents, peer-reviewed academic sources, and researcher-generated synthetic business decision scenarios. Optional expert review, if approved, will be limited to adults aged 18 or above with relevant expertise in digital transformation and emerging technologies, digital transformation, AI governance, smart-city management, Saudi policy, or Islamic business ethics. The project will not intentionally recruit children, patients, people lacking capacity to consent, pregnant women as a research category, economically dependent participants, or any group whose participation could create vulnerability or coercion. This exclusion is justified because the research question can be answered through documents, synthetic scenarios, and adult professional expertise.

9.3 Informed consent procedures

If optional expert participants are involved, each participant will receive a clear information sheet before participation. The sheet will explain the purpose of the study, what participation involves, expected duration, voluntary nature of participation, possible risks, expected benefits, data to be collected, storage procedures, confidentiality protections, withdrawal rights, and contact details for the researcher and supervisor. Consent will be obtained through a signed or university-approved electronic consent form. Participation will not be linked to grades, employment, institutional evaluation, financial reward, or any form of pressure. Participants may refuse to answer any question and may withdraw before anonymization or aggregation without penalty. This consent procedure operationalizes the principle of autonomy and the Belmont Report principle of respect for persons (U.S. Department of Health and Human Services, Office for Human Research Protections, n.d.-a).

9.4 Confidentiality, anonymity, and data minimization

The project will apply data minimization. Only information necessary to answer the research questions will be collected. No national identity numbers, passport numbers, personal addresses, private financial records, biometric data, health data, or sensitive personal data will be requested. If expert review is used, participants will be assigned codes such as E01, E02, and E03. Names, email addresses, and consent records will be stored separately from research notes and scoring files. Findings will be reported in aggregate form. Any quotation will be anonymized and stripped of institutional identifiers unless explicit permission is granted and ethics approval permits attribution. The project treats de-identification as risk reduction, not as an absolute guarantee, consistent with privacy-risk management logic (National Institute of Standards and Technology, 2020).

9.5 Data storage, access control, retention, and destruction

All research files will be stored in encrypted, password-protected folders under the researcher's control and, where required, in university-approved storage. Access will be limited to the researcher and supervisor. Files containing identifiers will be separated from analysis files. Where cloud storage is used, it will require strong authentication and will not be public or shared through unrestricted links. Data transfers will use encrypted

channels or university-approved systems. A source log, version log, and deletion log will be maintained. Identifiable data, if any, will be retained only for the period required by university policy and then securely deleted or destroyed. If no specific university retention rule applies, identifiable material will be destroyed after final assessment and administrative closure, while anonymized analysis outputs may be retained for auditability and future academic verification.

9.6 Legal and regulatory alignment

The proposal will follow applicable university rules and recognized data-protection principles. For Saudi-related data, it will respect the Saudi Personal Data Protection Law and its implementing regulations, especially lawful processing, purpose limitation, minimum necessary processing, confidentiality, and controller responsibility (Saudi Data and Artificial Intelligence Authority, n.d.-a; Saudi Data and Artificial Intelligence Authority, n.d.-b). Because the researcher is currently located in Europe, the project will also use GDPR-style principles—lawfulness, fairness, transparency, data minimization, storage limitation, integrity, and confidentiality—as a conservative compliance benchmark where relevant (European Parliament and Council of the European Union, 2016). This dual alignment controls risk; it does not imply that the project will process sensitive personal data or transfer operational data outside approved channels.

Table 9 consolidates the study’s ethical safeguards and data-protection procedures into an auditable protocol.

Table 9. Ethical Safeguards and Data-Protection Procedures

Ethical / data-protection area	Main risk	Control procedure	Residual risk / contingency
Institutional approval	Research begins before formal ethics clearance.	Submit full protocol, consent materials, instruments, and data plan to the relevant REC/IRB before any participant data collection.	If approval is delayed, proceed only with public-document analysis and synthetic scenario design.
Informed consent	Participants do not understand the study or feel pressured to participate.	Use a plain-language information sheet and signed/electronic consent; emphasize voluntariness, withdrawal rights, and refusal rights.	Exclude any participant whose consent is unclear or dependent on institutional pressure.
Confidentiality	Expert identity or institutional affiliation becomes inferable.	Use participant codes, aggregate reporting, paraphrasing where needed, and separate storage of identifiers.	Remove or generalize any quote or detail that creates re-identification risk.
Data minimization	Unnecessary personal or sensitive information is collected.	Collect only role-relevant expert feedback and no identity numbers, private financial data, health data, biometric data, or family data.	Delete unsolicited sensitive data immediately and document deletion in the audit log.
Storage and access	Unauthorized access, device loss, or accidental disclosure.	Use encrypted password-protected folders, restricted access, strong authentication, and separate storage of consent records.	Activate breach protocol: isolate files, notify supervisor/ethics authority, assess exposure, and follow institutional/legal reporting rules.
Retention and destruction	Data kept longer than necessary or destroyed without auditability.	Apply university retention rules; securely delete identifiers after administrative closure unless policy requires longer retention.	Maintain a deletion log and retain only anonymized analytical outputs where permissible.
Fair inclusion	Unjustified exclusion or recruitment of vulnerable groups.	Limit optional participants to adult experts whose expertise directly matches the study; exclude vulnerable groups because the research does not require them.	If a vulnerable participant is inadvertently approached, terminate recruitment and seek ethics guidance.

Negative-result integrity	Researcher manipulates coding or interpretation to force expected results.	Maintain codebook, source log, scenario assumptions, and scoring audit trail; report negative or mixed findings transparently.	Reframe contribution as boundary-condition evidence if the integrated framework does not outperform the baseline.
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Note. IRB = institutional review board. Expert review, if conducted, will not begin before ethics approval, exemption, or formal non-human-subjects determination. No confidential NEOM data, personal mobility data, or undisclosed institutional records will be used.

9.7 Risk-benefit assessment and ethical principles

The study presents minimal physical and psychological risk because it involves no experimental treatment, deception, minors, health data, or private operational data. The main risk is informational: accidental identification of an expert participant or unintended disclosure of coded feedback. The project controls that risk through minimization, coding, aggregation, secure storage, and restricted access. The expected benefit is academic and managerial: the study will produce a more disciplined way to evaluate AI-enabled digital transformation readiness in a Saudi smart-city context. The ethical design reflects four principles: autonomy, non-maleficence, beneficence, and justice. Autonomy is protected through informed consent and withdrawal rights; non-maleficence through confidentiality and low-risk design; beneficence through useful research outputs; and justice through fair inclusion criteria and avoidance of unnecessary vulnerable-group recruitment (U.S. Department of Health and Human Services, Office for Human Research Protections, n.d.-a).

9.8 Data breach and incident-response protocol

If a device, storage account, or research file is lost, stolen, or exposed, the researcher will immediately secure remaining accounts, change passwords, revoke shared access, document the incident, identify the affected files, and notify the supervisor and relevant ethics or data-protection authority according to university rules. If identifiable participant data are affected, the researcher will follow the applicable institutional and legal notification procedure. This protocol is designed to prevent a minor security incident from becoming an uncontrolled confidentiality breach.

If expert review is conducted, the participant information sheet, informed consent form, expert review form, and data-retention and destruction log will be used as shown in Appendices K-N. No expert input will be collected before institutional ethics approval, exemption, or non-human-subjects determination is documented.

10. LIMITATIONS AND DELIMITATIONS OF THE STUDY

This section separates the deliberate boundaries of the study from the methodological constraints that may affect interpretation. Delimitations define what the research intentionally includes and excludes. Limitations identify unavoidable weaknesses of the design and explain how they will be controlled. This distinction protects the project from overclaiming while showing that its scope has been selected for analytical depth and methodological feasibility. The distinction follows established research-writing guidance and methodological literature on limitations and delimitations (Theofanidis & Fountouki, 2018; University of Southern California Libraries, n.d.).

10.1 Delimitations of the Study

- The project is conceptually delimited to AI-enabled digital transformation readiness in Saudi smart-city megaprojects. It does not examine digital transformation across all industries, all public-sector institutions, or all urban development projects.

- The project uses NEOM as a bounded conceptual case setting, not as an object of operational audit. It does not evaluate NEOM's real performance, implementation success, financial results, citizen adoption, procurement execution, or institutional decision-making.
- The project is delimited to publicly available documents, peer-reviewed academic literature, coded assessment criteria, and synthetic NEOM-based business decision scenarios. It does not use confidential NEOM data, internal financial records, proprietary AI systems, personal mobility traces, or undisclosed government documents.
- The project focuses on assessment quality instead of real-world project performance. The dependent variable is strategic performance-readiness assessment quality, measured through analytical coverage, paired scenario scoring, coverage comparison, and sensitivity analysis.
- The project excludes macroeconomic impact, real citizen satisfaction, technical AI model accuracy, procurement performance, political feasibility, long-term financial return, and actual institutional adoption. These variables require longitudinal data, direct institutional access, or field-based evaluation beyond the scope of this digital-transformation proposal.

10.2 Limitations of the Study

- The first limitation is the use of synthetic business decision scenarios instead of real operational data. Synthetic scenarios cannot reproduce the full complexity of megaproject implementation, stakeholder behavior, procurement constraints, or institutional decision-making. The project will mitigate this limitation by documenting all scenario assumptions, using the same scenario set for both frameworks, and applying sensitivity analysis to test whether results remain stable under alternative scoring assumptions.
- The second limitation is limited external validity. Because NEOM is a distinctive Saudi smart-city megaproject, the findings may not generalize directly to conventional firms, older cities, small municipalities, or non-Saudi contexts. The project will therefore present its findings as context-specific and exploratory.
- The third limitation is criterion inflation. The integrated framework may appear stronger simply because it contains more dimensions than the generic checklist. The project will control this by distinguishing raw score improvement from analytical coverage gain, comparing overlapping and non-overlapping criteria, normalizing scoring procedures, and reporting whether added dimensions provide distinct audit value.
- The fourth limitation is coding subjectivity. Translating literature, policy requirements, AI governance principles, and Islamic business ethics into scoring criteria requires researcher judgment. This will be mitigated through a predefined codebook, explicit inclusion and exclusion rules, a source log, a transparent scoring rubric, and an audit trail.
- The fifth limitation concerns the operationalization of Islamic business ethics. Al-Amanah, Ihsan, and Maslaha may overlap with generic concepts such as accountability, quality management, and stakeholder value. The project will address this by translating each principle into auditable criteria and comparing its coverage against generic governance indicators.
- The sixth limitation is the absence of real institutional validation. Without access to NEOM decision-makers, internal project documents, or expert panels, the study cannot claim operational

validation. It will therefore describe its evaluation as preliminary internal validation under controlled synthetic scenario conditions.

- The seventh limitation is the possibility of negative or mixed findings. The integrated framework may not outperform the generic checklist across all dimensions. Such a result would not invalidate the research; it would identify the conditions under which integrated readiness frameworks add value, become redundant, or create excessive complexity.

Table 10 distinguishes deliberate scope boundaries from unavoidable methodological constraints and shows the mitigation strategy for each limitation.

Table 10. *Delimitations, Limitations, and Mitigation Controls*

Category	Scope or constraint	Mitigation / justification
Delimitation	NEOM is treated as a conceptual case setting rather than as an audited institution.	This protects the proposal from unsupported claims about real NEOM performance and keeps the study feasible without confidential access.
Delimitation	The study assesses framework quality, not actual implementation success.	The dependent variable remains analytical coverage and readiness-assessment quality, not real project outcomes.
Limitation	Synthetic scenarios cannot fully reproduce real-world smart-city complexity.	Scenario assumptions will be documented, paired scoring will be applied to the same cases, and sensitivity analysis will test robustness.
Limitation	The integrated framework may benefit from criterion inflation.	The analysis will separate raw scores from coverage gain, compare overlapping and non-overlapping criteria, and report whether added items create distinct analytical value.
Limitation	Single-researcher coding may introduce subjectivity.	A predefined codebook, audit trail, source log, and supervisor review where feasible will reduce arbitrary scoring.
Limitation	Islamic business-ethics criteria may overlap with generic governance concepts.	The study will use a coverage-overlap matrix to determine whether Al-Amanah, Ihsan, and Maslaha add distinguishable audit coverage.

Note. Delimitations are deliberate scope choices. Limitations are constraints that may affect interpretation. The table links each issue to a mitigation or justification.

10.3 Validity Position

Despite these limitations, the selected design remains appropriate for the research question. A full field study would require confidential data and institutional access that are not guaranteed at proposal stage. A purely conceptual study would not test whether the proposed framework improves assessment coverage. A purely financial model would ignore governance, stakeholder trust, and ethical legitimacy. Therefore, a controlled framework-building and synthetic-scenario evaluation design offers the strongest feasible approach for this digital-transformation proposal.

The project makes a deliberately narrow claim: it evaluates whether an integrated DTET-oriented framework provides broader and more traceable assessment coverage than a generic digital-transformation checklist under controlled Saudi smart-city scenario conditions. It does not claim to certify NEOM, predict implementation success, or measure national transformation outcomes.

11. CONTRIBUTION TO THE FIELD / SIGNIFICANCE AND IMPACT

This research offers a practical way to evaluate strategic performance readiness in AI-enabled digital transformation initiatives. After completion, reviewers will be able to compare a generic digital-transformation checklist with a Saudi-contextual DTET framework and identify the additional governance, finance, operations, and ethics coverage gained.

The theoretical contribution sits in digital transformation and emerging technologies and strategic digital transformation, not computer science as a whole. The project connects strategic management, digital transformation, AI governance, ROI logic, operational performance, stakeholder trust, Saudi policy, and Islamic ethical concepts. The practical contribution is a framework, rubric, checklist, synthetic scenario protocol, and managerial policy brief for managers, business analysts, smart-city planners, project evaluators, and governance officers.

The five-year vision is modest and realistic. The framework could become a first-generation template for future Saudi smart-city digital transformation governance research, later strengthened with real organizational data, expert panels, or project-level case studies. It will not certify NEOM systems, replace regulators, or claim to solve every digital transformation problem.

11.1 Expected Originality Statement

The expected original contribution of this digital-transformation research is the first-generation integration of digital transformation strategy, Saudi AI and data governance, performance-readiness assessment, and Islamic ethical criteria into one operational decision framework for Saudi smart-city megaproject contexts. Its originality does not depend on inventing an AI algorithm. It lies in making strategic readiness measurable, auditable, and context-specific in a setting where digital transformation, ROI, data governance, and ethics are usually treated as separate topics.

The added value is practical. Researchers and managers will be able to compare proposed AI-enabled initiatives through a documented rubric that links strategy, investment logic, operational performance, governance risk, stakeholder trust, and public-interest ethics.

11.2 Researcher Preparedness and Training Plan

The researcher is prepared to execute this project because the topic sits at the intersection of management, data analysis, financial-performance thinking, and supply-chain decision-making. Prior academic and project experience includes modelling decision problems, working with ROI, CAPEX and OPEX logic, using Excel, Power BI, Python, QGIS, and traffic-simulation concepts, and translating operational evidence into management recommendations. This background matters because the digital-transformation project requires analytical literacy and the ability to turn fragmented policy, ethical, and system-design evidence into an actionable managerial framework.

Remaining methodological development is treated as a formal training objective. During Months 1-2, the researcher will strengthen competence in strategic digital transformation literature, qualitative coding, MCDA scoring, research-data management, and research ethics. This plan supports feasibility without overstating current expertise.

The master's research period will also be used as an applied project environment, not merely as an individual dissertation exercise. Within supervisor guidance and university rules, the researcher will be ready to contribute to related applied research tasks, including literature mapping, data cleaning, governance-criteria coding, scoring-sheet design, documentation, presentation preparation, and project reporting.

12. RELEVANCE TO PRINCE SATTAM BIN ABDULAZIZ UNIVERSITY AND THE DTET PROGRAM

The research fits Prince Sattam bin Abdulaziz University's Digital Transformation and Emerging Technologies context because it focuses on AI-enabled transformation, emerging technologies, data governance, strategic performance readiness, investment discipline, risk control, stakeholder trust, and responsible implementation. It is framed as an applied digital-transformation problem, not a purely technical AI project.

The Study in Saudi application listing identifies Digital Transformation and Emerging Technologies as a master's-level program at Prince Sattam bin Abdulaziz University, delivered in English. The university's official public website also shows relevant institutional infrastructure, including graduate studies, IT, cybersecurity, research and development,

and quality/development structures. These signals support the fit between this proposal and a program focused on digital transformation, emerging technologies, governance, and applied performance evaluation.

Table 11 explains the fit between the proposed study, Prince Sattam bin Abdulaziz University’s digital-transformation program environment, and the resources required to execute the project.

Table 11. Departmental Fit and Resource Alignment

Fit dimension	Specific alignment	Resource or need	Why it matters
Project	Strategic governance of AI-enabled digital transformation initiatives.	DTET supervision in strategy, operations, finance, digital transformation, governance, or business analytics.	The question requires digital-transformation, emerging-technologies, governance, and performance-evaluation expertise rather than pure algorithm development.
People	Researcher will need literature synthesis, Excel/Python scoring, financial logic, and digital-transformation governance training.	Supervisor guidance; university workshops; open-source software.	Shows feasibility without overclaiming current expertise.
Place	Prince Sattam bin Abdulaziz University's graduate-program environment provides the digital-transformation and emerging-technologies setting.	Digital Transformation and Emerging Technologies program, business departments, quality/development structures, and academic evaluation process.	Explains why the project belongs in a digital-transformation program context.
Advisor fit	Final advisor should have expertise in strategy, digital transformation, operations, business analytics, governance, or performance management.	Final supervisor and research alignment to be inserted after formal assignment by the Graduate Studies.	Avoids empty name-dropping and requires actual methodological alignment.
Applied value	Framework can support decision-readiness assessment for digital initiatives.	Audit checklist, scoring matrix, managerial policy brief.	Connects academic research to managerial practice.

Note. The table links the proposal to digital-transformation and emerging-technologies fit, supervision needs, and methodological resources. It does not claim formal supervisor assignment.

13. RESEARCH PLAN, TIMELINE, AND DELIVERABLES

The project will be completed over a 12-month digital-transformation research period. The work plan is structured as a controlled project schedule, not a descriptive list of intentions. Each phase has a defined task, milestone, deliverable, dependency, and buffer. This matches project-scheduling logic, where a schedule model should support planning, execution, monitoring, closure, and delivery (Project Management Institute, 2011).

The timeline also protects ethical and methodological quality. If expert review or any human input is used, ethics screening, consent documentation, or an institutional determination will be completed before any contact or data collection. This sequencing protects the project from invalid data collection and avoidable IRB risk (U.S. Department of Health and Human Services, Office for Human Research Protections, n.d.-b).

13.1 Textual Gantt Chart

Table 12 turns the work plan into a controlled schedule by linking each phase to its milestone, deliverable, and dependency.

Table 12. Textual Gantt Chart: Twelve-Month Research Timeline and Deliverables

Phase	Months	Core activities	Milestone	Deliverable	Dependency / buffer
Phase 1: Scope confirmation and proposal control	1	Finalize title, problem statement, main research question, sub-questions, objectives, boundaries, and source strategy.	Approved research scope.	Final research design map and source-search protocol.	No major variable or research question is added after scope lock.
Phase 2: Literature review and source matrix	1-3	Review strategic management, digital transformation, AI	Core literature base completed.	Thematic literature matrix with 15-20 core sources.	Month 3 includes buffer for source access and matrix revision.

Phase	Months	Core activities	Milestone	Deliverable	Dependency / buffer
		governance, performance measurement, Saudi smart-city context, and Islamic business ethics.			
Phase 3: Ethics screening and data-protection clearance	2-3	Complete ethics screening / IRB determination if expert review or human input is used; prepare informed-consent materials if needed.	Ethics route confirmed.	Ethics decision record and consent template if applicable.	No expert contact or human input before clearance.
Phase 4: Codebook and variable operationalization	3-5	Build codebook, define variables, construct 0-3 scoring logic, and map theory to assessment dimensions.	Codebook frozen.	Variable table, scoring rubric, and theoretical mapping table.	Month 5 includes buffer for supervisor feedback.
Phase 5: Ethics mapping and coverage logic	5-6	Translate Al-Amanah, Ihsan, and Maslaha into auditable managerial criteria and compare overlap with generic governance indicators.	Ethics criteria operationalized.	Islamic ethics mapping table and coverage-overlap logic.	Must be completed before full scenario scoring.
Phase 6: Pilot scoring test	6	Test the rubric on 3-5 synthetic scenarios; detect unclear criteria, scoring ambiguity, and criterion inflation.	Pilot completed.	Revised rubric v2.	Mandatory quality gate before full scenario generation.
Phase 7: Synthetic scenario construction	6-7	Build at least 30 synthetic NEOM-based business decision scenarios with fixed assumptions and documented variables.	Scenario set finalized.	Scenario database / spreadsheet and assumptions log.	Buffer included for scenario redesign if pilot fails.
Phase 8: Framework evaluation	8-9	Apply the generic checklist and integrated DTET framework to the same scenario set; conduct paired scoring and coverage comparison.	Comparative scoring completed.	Paired scoring table and coverage-gain matrix.	Same scenarios and same scoring scale used for both frameworks.
Phase 9: Sensitivity and robustness checks	9	Test whether results change under alternative weights, normalized scoring, or removal of overlapping criteria.	Robustness check completed.	Sensitivity-analysis table.	Protects against criterion inflation and scoring bias.
Phase 10: Findings, limitations, and discussion	10	Interpret results, identify negative or mixed findings, and write validity position, limitations, delimitations, and risk implications.	Results chapter drafted.	Findings and discussion draft.	Buffer for weak or negative findings.
Phase 11: Full dissertation writing	10-11	Integrate all chapters: introduction, literature review, methodology, framework, results, discussion, contribution, ethics, and limitations.	Full draft completed.	Complete supervisor-review draft.	Month 11 reserved for supervisor feedback.
Phase 12: Revision, formatting, and defense preparation	12	Final corrections, reference audit, formatting, appendices, presentation preparation, submission, and defense rehearsal.	Final submission ready.	Final document, slides, and defense notes.	Final two weeks reserved as emergency buffer.

Note. DTET = Digital Transformation and Emerging Technologies. The table sequences the study by phases, milestones, deliverables, dependencies, and buffers across a 12-month research cycle.

Figure 3 functions as a project-control mechanism by linking each research phase to a milestone, deliverable, quality gate, and buffer period.

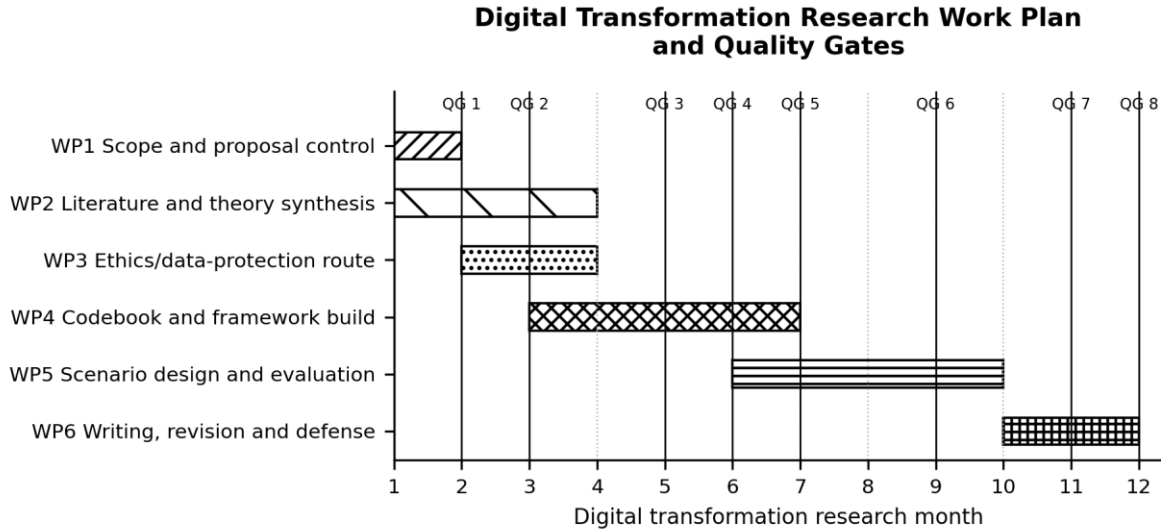


Figure 3. Twelve-Month Work-Package Gantt Chart Linking Research Phases, Deliverables, Quality Gates, and Buffers

Note. WP = work package; QG = quality gate; DTET = Digital Transformation and Emerging Technologies. The chart covers a 12-month digital-transformation research cycle and sequences the project from scope confirmation and literature review to ethics screening, codebook development, pilot scoring, synthetic scenario evaluation, robustness checks, writing, revision, and submission. Buffer periods are included after literature review, ethics screening, pilot scoring, evaluation, and final formatting. The figure is a project-control tool, not an empirical result.

13.2 Work-Package Summary

Table 13 condenses the timeline into work packages so that the project can be monitored by output instead of by intention.

Table 13. Work-Package Summary

Work package	Timeframe	Purpose	Output
WP1. Proposal control and research design	Month 1	Lock the research problem, questions, objectives, scope, and boundaries.	Research design map.
WP2. Literature and theoretical synthesis	Months 1-3	Build the academic and theoretical base for the framework.	Literature matrix and theoretical framework.
WP3. Ethics and data-protection route	Months 2-3	Confirm ethics requirements and prepare consent/data-protection materials if needed.	Ethics screening record and data-protection protocol.
WP4. Framework construction	Months 3-6	Build the codebook, variables, scoring rubric, ethics mapping, and pilot test.	Pilot-tested integrated readiness framework.
WP5. Scenario design and evaluation	Months 6-9	Create synthetic scenarios, compare frameworks, and test robustness.	Scenario set, paired scoring, coverage comparison, sensitivity analysis.
WP6. Writing, revision, and defense	Months 10-12	Write, revise, format, submit, and defend the final document.	Final submission and defense materials.

Note. WP = work package. The table compresses the research plan into executable work packages with named outputs.

13.3 Quality Gates

Table 14 defines the quality gates that must be passed before the project moves to the next major stage.

Table 14. Quality Gates and Pass Conditions

Quality gate	Deadline	Pass condition
QG1. Scope lock	End of Month 1	No new research questions or major variables added after this point.
QG2. Ethics route confirmed	End of Month 3	Ethics approval, exemption, or non-human-subjects determination documented before any expert input.
QG3. Codebook freeze	End of Month 5	Coding categories and scoring scale approved for pilot testing.
QG4. Pilot scoring pass	End of Month 6	Rubric scores scenarios without major ambiguity or unmanageable criterion overlap.

Quality gate	Deadline	Pass condition
QG5. Scenario set lock	End of Month 7	Scenario assumptions fixed before comparative scoring.
QG6. Robustness check completed	End of Month 9	Sensitivity analysis completed before conclusions are written.
QG7. Full draft completed	End of Month 11	Complete document ready for supervisor review.
QG8. Final submission package	Month 12	Formatting, references, appendices, and defense materials complete.

Note. QG = quality gate. Each gate defines a pass condition that must be met before the next major research stage proceeds.

13.4 Buffer Strategy

Table 15 shows how schedule buffers are placed at points where research delay or revision risk is most likely.

Table 15. Buffer Strategy and Schedule-Risk Controls

Buffer	Placement	Purpose
Literature buffer	Month 3	Absorb delays in source access, matrix completion, and reference screening.
Ethics buffer	Months 2-3	Allow time for ethics screening or confirmation that no human-subjects review is required.
Pilot buffer	Month 6	Correct weak scoring criteria before full scenario evaluation.
Evaluation buffer	Month 9	Handle sensitivity analysis, weighting changes, or criterion-inflation checks.
Final buffer	Month 12	Handle supervisor feedback, formatting, references, appendices, and defense preparation.

Note. The buffers are schedule-risk controls, not unused time. They are placed after stages where delay or revision risk is highest.

The minimum viable design is protected if delays occur: literature matrix, codebook, integrated framework, synthetic scenarios, paired scoring, coverage comparison, and sensitivity analysis.

Optional outputs such as a longer policy brief, external expert review, or publication preparation can be reduced or postponed without damaging the core research question.

13.5 Dissemination Cascade

The dissemination strategy is set before dissertation writing begins. The project will not be submitted to multiple venues at the same time. It will follow a sequential cascade: the strongest fit comes first; if rejected, the manuscript will be reformatted for the next venue after supervisor approval. The objective remains realistic for an digital-transformation project: one conference abstract or poster and one draft journal-style manuscript, not a guaranteed Q1 publication.

Table 16 frames dissemination as a sequential cascade and prevents the proposal from overpromising publication outcomes.

Table 16. Dissemination Cascade

Priority	Target venue	Fit with study	Planned output and timing
A	DTET / digital-transformation research seminar or postgraduate symposium	Best fit for a framework-building and managerial-readiness study.	Short presentation after Month 11 if results are coherent.
B	Digital transformation / business analytics conference track	Matches AI-enabled transformation, readiness, and managerial evaluation.	Conference abstract after dissertation submission.
C	Responsible AI / governance workshop	Fits AI governance, accountability, and assessment coverage.	Poster or short paper if governance findings are strong.
D	Smart-city management or public-value venue	Fits Saudi smart-city context and stakeholder-value framing.	Post-master's extension with supervisor approval.
E	Journal-style manuscript draft	Longer-term output only if the framework and results are sufficiently robust.	Prepared after final defense, not promised as guaranteed publication.

Note. The dissemination cascade is a planning tool, not a promise of publication acceptance. Final venue selection will depend on supervisor guidance and actual results.

The cascade is a planning tool, not a promise of acceptance. The supervisor should revise it once final scenario-based results are available.

13.6 DTET Proposal Evaluation Readiness Audit

Table 17 functions as a final proposal audit by mapping evaluator concerns to the sections and evidence that address them.

Table 17. DTET Proposal Evaluation Readiness Audit

Evaluation area	Where addressed	Evidence of readiness	Residual risk control
Title and cover page	Cover page	Clear DTET framing, Saudi smart-city context, NEOM case setting, and digital-transformation and emerging-technologies keywords.	Replace pending administrative fields before submission.
Introduction and background	Sections 1-2	Territory, gap, boundaries, and program relevance are stated.	Avoid promotional language about NEOM.
Problem and research questions	Section 3	One main research question and four aligned sub-questions.	Maintain objective wording; do not presume framework superiority.
Aim and objectives	Section 4	One main aim and four SMART objectives.	Do not add new objectives after scope lock.
Literature and theory	Sections 5-6	Thematic synthesis, theoretical mapping, and propositions.	Avoid name-dropping; keep each theory linked to variables.
Methodology and validation	Sections 7-8	Mixed-methods case logic, rubric, paired scoring, sensitivity analysis, and risk register.	Frame findings as preliminary internal validation.
Ethics and data protection	Section 9	Consent logic, data minimization, storage, retention, destruction, and breach response.	No expert contact before ethics clearance if required.
Limitations and delimitations	Section 10	Scope boundaries and limitations are separated and mitigated.	Do not present absence of real NEOM data as operational validation.
Timeline and feasibility	Section 13	Gantt structure, quality gates, buffers, and minimum viable design.	Cut optional outputs first if delays occur.
References and appendices	Section 14 and appendices	Reference audit, scenario schema, codebook, and response template.	Final reference-manager check before submission.

Note. The audit table maps proposal sections to evaluator concerns and residual risk controls. It is intended for self-review before submission.

13.7 Reference Quality Audit and Citation Control

Table 18 defines citation-control rules that will be used before final submission to reduce reference-list errors.

Table 18. Reference Quality Audit and Citation Control

Audit rule	Implementation in final submission
Sequential numbering	Citations appear in the order of first use.
One-to-one correlation	Every cited source appears in the reference list; every listed source is cited in the body.
Reference manager	Zotero, Mendeley, or a verified manual audit will be used to reduce citation error.
Metadata completeness	Author names, title, year, venue, DOI or URL, and access date are checked where applicable.
Citation flow	Citations support background, literature review, methodology, validation logic, legal claims, and scheduling logic.
Source hierarchy	Peer-reviewed articles and official institutional documents are prioritized over blogs or generic web pages.
URL hygiene	Tracking parameters and unnecessary URL fragments are removed from the final reference list.

Note. The audit rules are designed to keep citations sequential, traceable, complete, and consistent with the final reference list.

13.8 Reviewer-Response Preparedness

If reviewers comment on the proposal, the revision process will use a point-by-point response protocol. The researcher will answer each comment directly, state the revision made, or give a concise evidence-based reason for declining a change. Appendix O provides the response template.

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APPENDICES

The appendices contain only operational instruments needed to audit, reproduce, or ethically review the proposed study. They do not introduce new methods or arguments beyond those explained in the main text.

Appendix	Content	Function in the proposal	Main-text callout
A	Source selection protocol	Controls source quality and inclusion logic	Methodology / Literature Review
B	Literature review matrix	Documents synthesis evidence without overloading the body	Literature Review
C	Coding protocol and codebook	Supports reproducible document analysis	Methodology
D	Integrated readiness rubric	Defines 0-3 scoring criteria	Validation / Methodology
E	Islamic ethics mapping	Tests whether ethics criteria add coverage beyond generic governance	Ethics / Framework
F	Synthetic scenario protocol	Defines scenario construction rules	Methodology

G	Scenario bank and data schema	Allows audit of scenario variables	Methodology
H	Baseline checklist	Documents the comparator instrument	Validation
I	Paired scoring template	Supports baseline vs integrated framework comparison	Data analysis
J	Sensitivity analysis template	Controls criterion inflation and weighting bias	Risk / Analysis
K-N	Ethics and data-protection instruments	Support IRB/DPO review if expert review is conducted	Ethics
O	Reviewer-response template	Supports structured revision process	Reviewer-response preparedness

Appendix A. Source Selection Protocol and Inclusion/Exclusion Criteria

Purpose: This appendix defines how sources will be selected, screened, and excluded so that the literature review and policy analysis remain transparent and reproducible.

Element	Protocol
Source types	Peer-reviewed journal articles; official Saudi policy and university sources; recognized AI governance frameworks; business-administration and strategy literature.
Inclusion criteria	Direct relevance to digital transformation, AI governance, strategic readiness, performance measurement, smart-city governance, Saudi context, or Islamic business ethics.
Exclusion criteria	Opinion-only pieces, promotional material without analytical value, outdated sources unless seminal, duplicate sources, and sources not linked to variables or methods.
Screening output	Source log with citation, source type, theme, contribution, limitation, and relevance code.

Appendix B. Literature Review Matrix

Purpose: This appendix records the extended literature synthesis without turning the main literature review into a descriptive catalogue.

Field	Content to record
Citation	Author, year, title, venue, DOI/URL if applicable.
Theme	Strategic management, digital transformation, AI governance, performance measurement, Saudi smart-city context, or Islamic business ethics.
Contribution	What the source adds to the framework.
Limitation	What the source does not solve.
Link to study	Relevant variable, dimension, or methodological choice.

Appendix C. Coding Protocol and Codebook

Purpose: This appendix defines coding families and rules for extracting requirements from literature and public documents.

Code family	Evidence rule
STRATEGY_ALIGNMENT	Evidence must relate to strategic fit, capability logic, value creation, or transformation priorities.
INVESTMENT_DISCIPLINE	Evidence must relate to ROI logic, cost justification, resource allocation, or risk-adjusted value.
OPERATIONAL_FEASIBILITY	Evidence must relate to processes, implementation capacity, skills, monitoring, or execution risk.
GOVERNANCE_RISK	Evidence must relate to accountability, transparency, auditability, oversight, data governance, or escalation controls.
STAKEHOLDER_TRUST	Evidence must relate to legitimacy, fairness, stakeholder impact, communication, or public-value expectations.
ISLAMIC_ETHICS	Evidence must relate to Al-Amanah, Ihsan, Maslaha, stewardship, excellence, or public interest.

Appendix D. Integrated DTET Readiness Framework Rubric

Purpose: This appendix provides the scoring logic used to evaluate each synthetic business decision scenario.

Score	Meaning
0	Absent: the criterion is not addressed.
1	Weak: the criterion is mentioned but not operationalized.
2	Adequate: the criterion is operationalized with partial evidence or controls.
3	Strong: the criterion is clearly operationalized, traceable, and linked to managerial decision-making.
Composite logic	Scores will be reported by dimension and as normalized analytical coverage, not as proof of real project performance.

Appendix E. Islamic Business Ethics Mapping Table

Purpose: This appendix prevents Islamic business ethics from becoming decorative by translating each concept into auditable criteria.

Principle	Operational meaning	Example audit item
Al-Amanah	Trustworthy stewardship and accountability	Roles, data responsibility, documentation, and audit trails are explicitly assigned.
Ihsan	Excellence, quality, and continuous improvement	Monitoring, correction, learning, and quality-control mechanisms are specified.
Maslaha	Public interest and harm reduction	Stakeholder benefit, accessibility, risk reduction, and public-value justification are documented.
Coverage test	Distinctiveness check	Criteria will be compared against generic governance indicators to identify overlap and unique contribution.

Appendix F. Synthetic Scenario Protocol

Purpose: This appendix defines how synthetic NEOM-based business decision scenarios will be constructed without using confidential or personal data.

Protocol item	Rule
Scenario unit	One AI-enabled digital transformation initiative framed as a managerial decision scenario.
Scenario count	Target minimum: 30 scenarios; smaller pilot only if justified by supervisor-approved constraints.
Variation rules	Scenarios vary by strategic alignment, investment discipline, operational feasibility, governance risk, stakeholder trust, and ethical legitimacy.
Fixed assumptions	NEOM is used as a bounded conceptual case setting; no real NEOM operational data are claimed.
Documentation	Each scenario includes assumptions, variable values, scoring notes, and limitations.

Appendix G. Synthetic Scenario Bank and Data Schema

Purpose: This appendix lists the planned fields used to build and audit the synthetic scenario bank.

Field	Type	Purpose
scenario_id	Integer	Unique scenario identifier.
initiative_type	Categorical	AI-enabled service, data platform, mobility support, operational dashboard, or customer-facing service.
strategy_fit	Ordinal 0-3	Level of strategic alignment.
investment_logic	Ordinal 0-3	Strength of ROI and cost justification.
operational_feasibility	Ordinal 0-3	Implementation feasibility.
governance_risk	Ordinal 0-3	Strength of governance controls.
stakeholder_trust	Ordinal 0-3	Trust and legitimacy coverage.
islamic_ethics_coverage	Ordinal 0-3	Coverage of Al-Amanah, Ihsan, and Maslaha.

Appendix H. Generic Digital-Transformation Checklist Used as Baseline

Purpose: This appendix documents the baseline comparator so that the integrated framework is not compared against an undefined or artificially weak instrument.

Baseline dimension	Generic checklist focus
Technology readiness	Availability of digital tools and infrastructure.
Process readiness	Basic process digitalization and workflow adjustment.
People readiness	Skills, training, and adoption capacity.
Data readiness	Data availability, basic quality, and accessibility.
Performance readiness	Expected benefits and generic KPI logic.
Limitation	The baseline does not fully integrate Saudi smart-city context, Islamic business ethics, or detailed governance-risk controls.

Appendix I. Paired Scoring Sheet Template

Purpose: This appendix defines the comparison sheet used to score the same scenario under both frameworks.

Column	Description
scenario_id	Unique scenario number.
baseline_score	Score from generic digital-transformation checklist.
integrated_score	Score from integrated digital-transformation readiness framework.
normalized_difference	Difference after normalization.
coverage_gain	Additional non-overlapping analytical coverage.
scoring_notes	Brief rationale for scoring decision.

Appendix J. Sensitivity Analysis Template

Purpose: This appendix protects the study from criterion inflation and weighting bias.

Test	Purpose
Equal-weight model	Checks results when all dimensions receive equal weight.
Normalized-score model	Prevents the integrated framework from winning only because it has more criteria.
Overlap-removal test	Removes criteria duplicated in both frameworks and retests coverage gain.
Weight-shift test	Tests whether conclusions survive alternative dimension weights.
Negative-result protocol	If the advantage disappears, the study reports boundary conditions rather than claiming superiority.

Appendix K. Participant Information Sheet

Purpose: This appendix will be used only if expert review or human input is conducted.

Required item	Content
Study title	Full proposal title.
Researcher identity	Name, program, university, and contact details.
Purpose of participation	Review of framework clarity, relevance, and feasibility.
Voluntary participation	Participation is voluntary and may be withdrawn without penalty.
Risks and benefits	Minimal risk; contribution to research design quality.
Confidentiality	Responses will be anonymized or reported in aggregate unless explicit permission is granted.

Appendix L. Informed Consent Form

Purpose: This appendix records voluntary consent if expert review or any human-subject input is included.

Consent statement	Participant confirms
I have read the information sheet.	Yes / No
I understand the purpose of the study.	Yes / No
I understand that participation is voluntary.	Yes / No
I understand that I may withdraw before data analysis begins.	Yes / No
I consent to anonymized use of my responses.	Yes / No
Signature/date if required	Participant / researcher record.

Appendix M. Expert Review Form / Semi-Structured Review Guide

Purpose: This appendix defines the optional expert-review instrument without introducing intrusive or unnecessary questions.

Review area	Question
Clarity	Are the framework dimensions understandable and non-overlapping?
Relevance	Do the dimensions reflect DTET-level decision-making needs?
Feasibility	Can the rubric be applied within the proposed 12-month timeline?
Completeness	Which critical readiness dimension is missing, if any?
Scoring usability	Are any 0-3 scoring criteria ambiguous or hard to apply?
Final comment	What revision would most improve the framework?

Appendix N. Data Management, Retention, and Destruction Log

Purpose: This appendix documents how research materials will be stored, protected, retained, and destroyed.

Data type	Storage/access	Retention/destruction
Source log and literature matrix	Encrypted folder; researcher and supervisor access only	Retain until final approval, then archive according to university rules.
Synthetic scenarios and scoring sheets	Encrypted folder with version control	Retain for reproducibility unless supervisor requires deletion.
Expert-review data if collected	Separate encrypted folder; de-identified file names	Destroy identifiable data after approved retention period.
Consent records if used	Separate protected folder from research responses	Retain only as required by ethics approval.
Breach response	Notify supervisor and ethics contact immediately	Document incident, contain access, and follow institutional procedure.

Appendix O. Reviewer-Response Template

Purpose: This appendix supports disciplined revision after supervisor, committee, or reviewer comments.

Reviewer comment	Response strategy	Revision made	Location in revised document
Comment 1	Accept, partially accept, or decline with evidence.	State exact change.	Section/page/paragraph/table.
Comment 2	Clarify misunderstanding if needed.	Add explanation or revise wording.	Section/page/paragraph/table.
Comment 3	Resolve conflicting feedback through supervisor guidance.	State selected direction.	Section/page/paragraph/table.
Comment 4	If declined, justify with method or citation.	No change or partial clarification.	Section/page/paragraph/table.